

2013
**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Chemistry

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet and a Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 13, 17, 19, 21 and 25

Total marks – 100

Section I Pages 2–28

75 marks

This section has two parts, Part A and Part B

Part A – 20 marks

- Attempt Questions 1–20
- Allow about 35 minutes for this part

Part B – 55 marks

- Attempt Questions 21–31
- Allow about 1 hour and 40 minutes for this part

Section II Pages 29–44

25 marks

- Attempt ONE question from Questions 32–36
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 20 marks

Attempt Questions 1–20

Allow about 35 minutes for this part

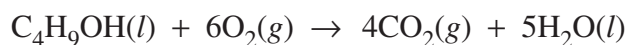
Use the multiple-choice answer sheet for Questions 1–20.

- 1 Which pair of reactants is used to industrially synthesise ammonia?
 - (A) H_2 and C
 - (B) H_2 and N_2
 - (C) H_2O and N_2
 - (D) H_2O and NO_2

- 2 What is the purpose of the flame in atomic absorption spectroscopy (AAS)?
 - (A) To ionise the sample
 - (B) To produce a spectrum
 - (C) To atomise the substance
 - (D) To provide the absorption wavelength

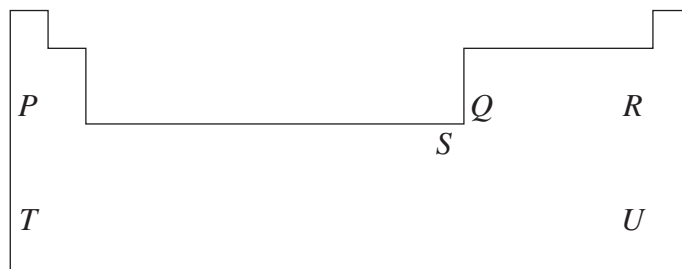
- 3 What is the main environmental problem associated with chlorofluorocarbon compounds?
 - (A) Acid rain
 - (B) Eutrophication
 - (C) Global warming
 - (D) Ozone depletion

- 4 Butan-1-ol burns in oxygen according to the following equation.



How many moles of carbon dioxide would form if two moles of butan-1-ol were burnt in excess oxygen?

- (A) 2
(B) 4
(C) 8
(D) 10
- 5 When placed in the Periodic Table, the recently discovered element 116 would be found in the same group as
- (A) element 16.
(B) element 43.
(C) element 87.
(D) element 102.
- 6 A representation of the Periodic Table is shown. The positions of six different elements, *P*, *Q*, *R*, *S*, *T* and *U* are given.



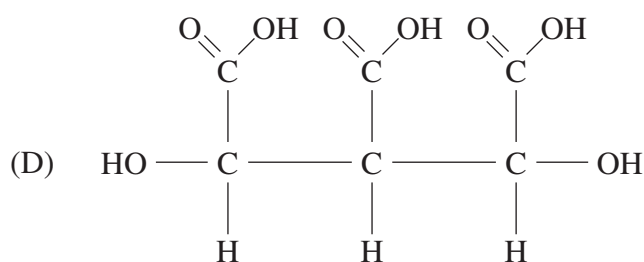
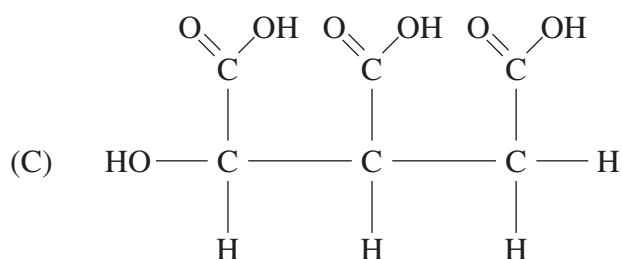
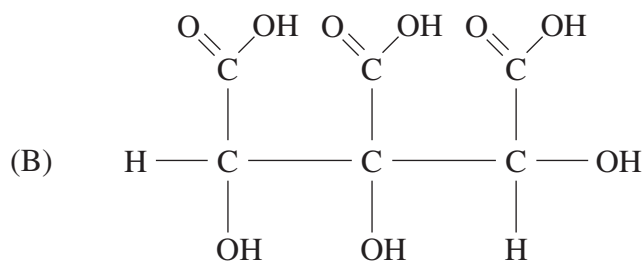
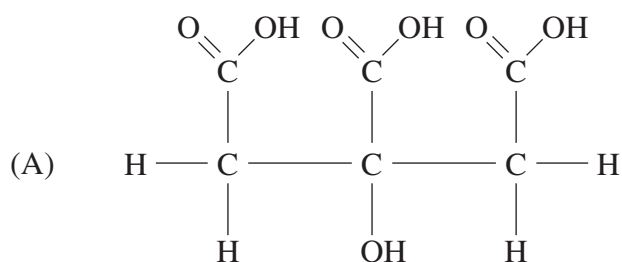
Which row of the following table shows the correct acid and base reactivities of the oxides of these elements?

	<i>Oxide reacts with acid only</i>	<i>Oxide reacts with base only</i>	<i>Oxide reacts with both acid and base</i>
(A)	<i>P, T</i>	<i>R, U</i>	<i>Q, S</i>
(B)	<i>P, R</i>	<i>T, U</i>	<i>Q, S</i>
(C)	<i>Q, R</i>	<i>P, S</i>	<i>T, U</i>
(D)	<i>Q, S</i>	<i>R, U</i>	<i>P, T</i>

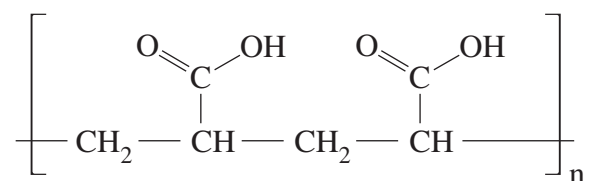
7 Which of the following is the correct equation for an INCOMPLETE combustion of octane?

- (A) $\text{C}_8\text{H}_{18}(l) + 8\text{O}_2(g) \rightarrow 8\text{CO}_2(g) + 9\text{H}_2(g)$
 (B) $\text{C}_8\text{H}_{18}(l) + \frac{25}{2}\text{O}_2(g) \rightarrow 8\text{CO}_2(g) + 9\text{H}_2\text{O}(l)$
 (C) $\text{C}_8\text{H}_{18}(l) + 6\text{O}_2(g) \rightarrow 3\text{CO}(g) + 5\text{C}(s) + 9\text{H}_2\text{O}(l)$
 (D) $\text{C}_8\text{H}_{18}(l) + 13\text{O}_2(g) \rightarrow 4\text{CO}(g) + 4\text{C}(s) + 9\text{H}_2\text{O}(l)$

8 Which of the following structural formulae shows citric acid, also known as 2-hydroxypropane-1,2,3-tricarboxylic acid?



- 9 A portion of a resin made from acrylic acid ($\text{CH}_2=\text{CHCOOH}$) is shown.



Which type of reaction results in the formation of this polymer?

- (A) Addition
 - (B) Condensation
 - (C) Dehydration
 - (D) Esterification
- 10 The following equilibrium is set up in a sealed reaction vessel.



Which of the following would INCREASE the yield of nitrogen dioxide?

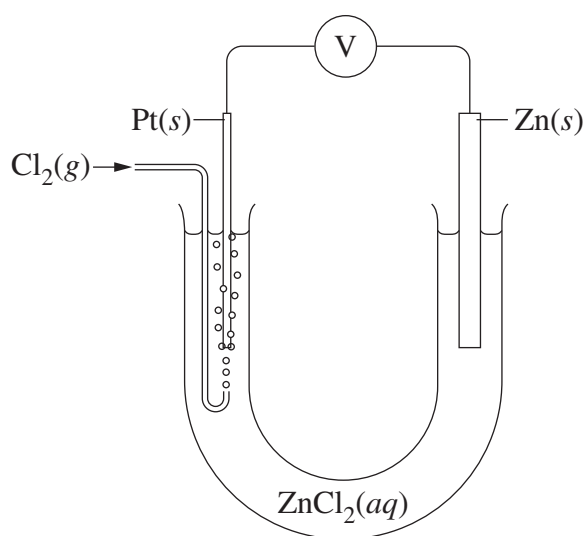
- (A) Adding a catalyst to the reaction vessel
 - (B) Decreasing the volume of the reaction vessel
 - (C) Raising the temperature of the reaction vessel
 - (D) Increasing the pressure by adding argon to the reaction vessel
- 11 The table shows the heat of combustion for four compounds.

<i>Compound</i>	<i>Heat of combustion</i> (kJ mol^{-1})
CO	233
CH_4	890
C_2H_2	1300
C_2H_6	1560

Which of these compounds would produce the greatest amount of energy if 1.00 g of each is burnt?

- (A) CO
- (B) CH_4
- (C) C_2H_2
- (D) C_2H_6

12 An experiment was set up as shown.

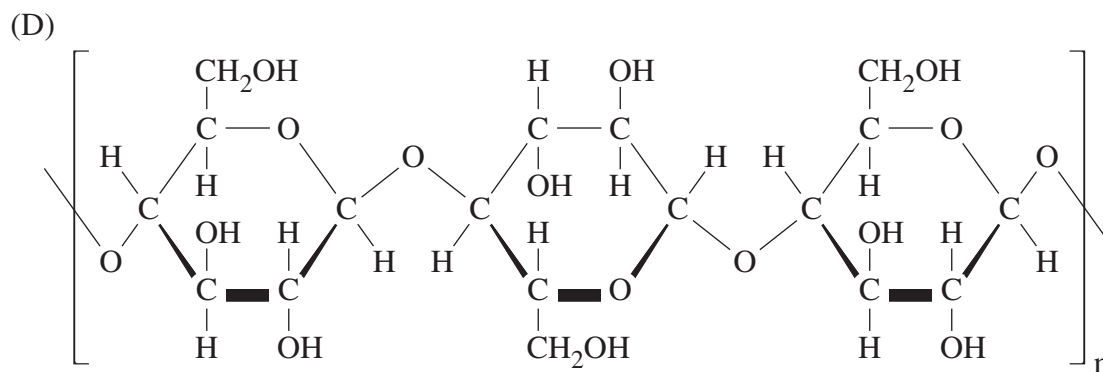
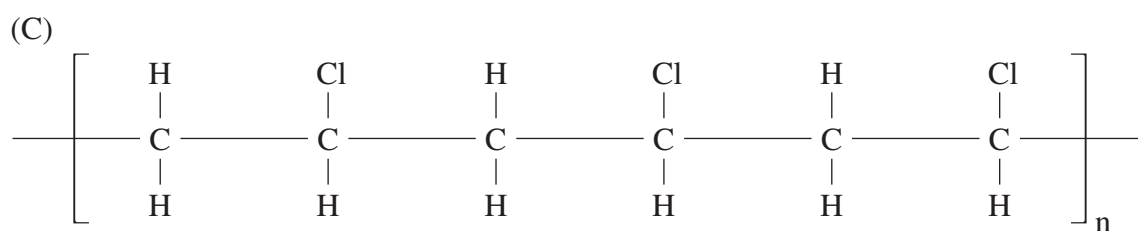
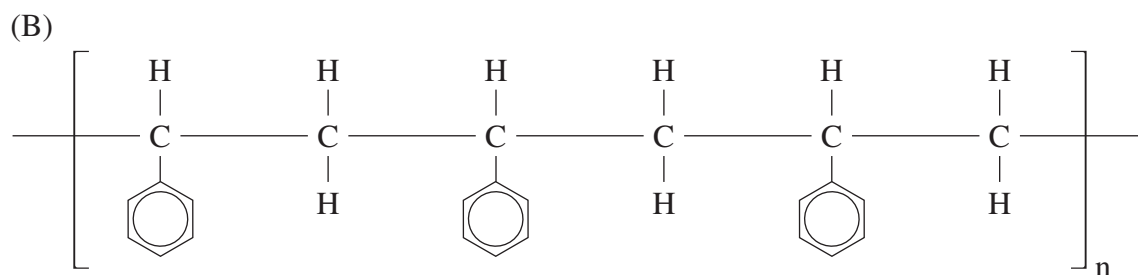
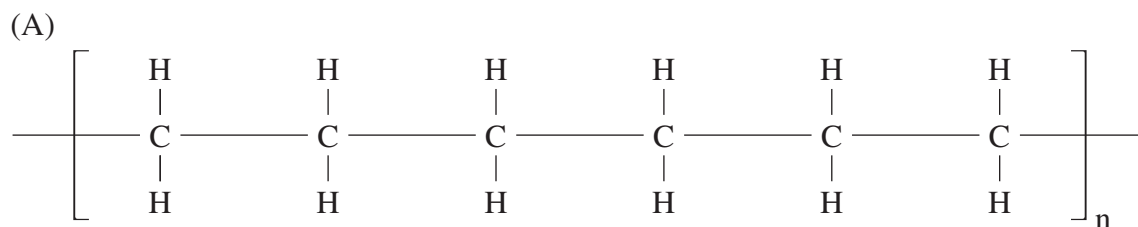


Which of the following statements is correct?

- (A) The chlorine gas is the anode.
- (B) The zinc electrode is the anode.
- (C) The platinum electrode is the anode.
- (D) There is no anode because there is no salt bridge.



What is the structure of the polymer most likely to have been used in the manufacture of the cup shown?



- 14** Sodium reacts with water to give hydrogen gas and sodium hydroxide solution.

What volume of gas would be produced from the reaction of 22.99 g of sodium at 25°C and 100 kPa?

- (A) 11.36 L
- (B) 12.40 L
- (C) 22.71 L
- (D) 24.79 L

- 15** A pH 3.0 solution of $\text{HCl}(aq)$ is diluted by adding water to produce a pH 5.0 solution.

Which row in the following table correctly identifies an appropriate volume of the original solution and the volume of water added for this dilution?

	<i>Volume of original solution (mL)</i>	<i>Volume of water added (mL)</i>
(A)	100	900
(B)	100	1000
(C)	10	990
(D)	1	1000

- 16** Which two species will react to form a product containing a coordinate covalent bond?

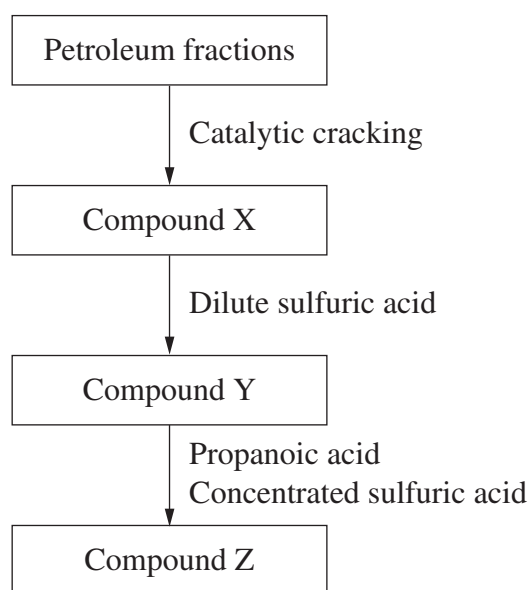
- (A) $\text{Ca}(s)$ and $2\text{H}^+(aq)$
- (B) $\text{H}_2\text{O}(l)$ and $\text{H}^+(aq)$
- (C) $\text{Ag}^+(aq)$ and $\text{Cl}^-(aq)$
- (D) $\text{NH}_4^+(aq)$ and $\text{OH}^-(aq)$

- 17** A 25.0 mL sample of a 0.100 mol L^{-1} hydrochloric acid solution completely reacted with 23.4 mL of sodium hydroxide solution.

What volume of the same sodium hydroxide solution would be required to completely react with 25.0 mL of a 0.100 mol L^{-1} acetic acid solution?

- (A) Less than 23.4 mL
- (B) 23.4 mL
- (C) More than 23.4 mL
- (D) Unable to calculate unless the concentration of the sodium hydroxide solution is also known

18 Consider the following series of reactions.



Which row in the table correctly identifies Compounds X, Y and Z?

	<i>Compound X</i>	<i>Compound Y</i>	<i>Compound Z</i>
(A)	Propene	Propan-1-ol	Ethyl propanoate
(B)	Propene	Ethanol	Propyl ethanoate
(C)	Ethanol	Ethylene	Propyl ethanoate
(D)	Ethylene	Ethanol	Ethyl propanoate

- 19 A solution was obtained by boiling flowers in water. After various substances were added to separate samples of the solution, the colour of each was noted.

<i>Substance added</i>	<i>Colour observed</i>
$0.1 \text{ mol L}^{-1} \text{ HCl}(aq)$	Bright pink
$0.01 \text{ mol L}^{-1} \text{ HCl}(aq)$	Bright pink
$0.001 \text{ mol L}^{-1} \text{ HCl}(aq)$	Pale yellow
Distilled water	Bright yellow
$0.001 \text{ mol L}^{-1} \text{ NaOH}(aq)$	Bright yellow
$0.01 \text{ mol L}^{-1} \text{ NaOH}(aq)$	Bright yellow

For which of the following titrations would it be appropriate to use this solution as an indicator?

- (A) $\text{HCl}(aq) + \text{NH}_3(aq)$
(B) $\text{HCl}(aq) + \text{NaOH}(aq)$
(C) $\text{CH}_3\text{COOH}(aq) + \text{NH}_3(aq)$
(D) $\text{CH}_3\text{COOH}(aq) + \text{NaOH}(aq)$
- 20 The structures of ozone and molecular oxygen are shown.



Ozone is more easily decomposed than molecular oxygen because

- (A) it is polar.
(B) it is a bent molecule.
(C) it has a greater molecular mass.
(D) it has a lower average bond energy.

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Centre Number

Section I (continued)

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Part B – 55 marks

Attempt Questions 21–31

Student Number

Allow about 1 hour and 40 minutes for this part

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Show all relevant working in questions involving calculations.

Question 21 (6 marks)

Please turn over

Question 21 (6 marks)

An ester and an acid are both listed as additives on a food label.

(a) Why is acid used as a food preservative?

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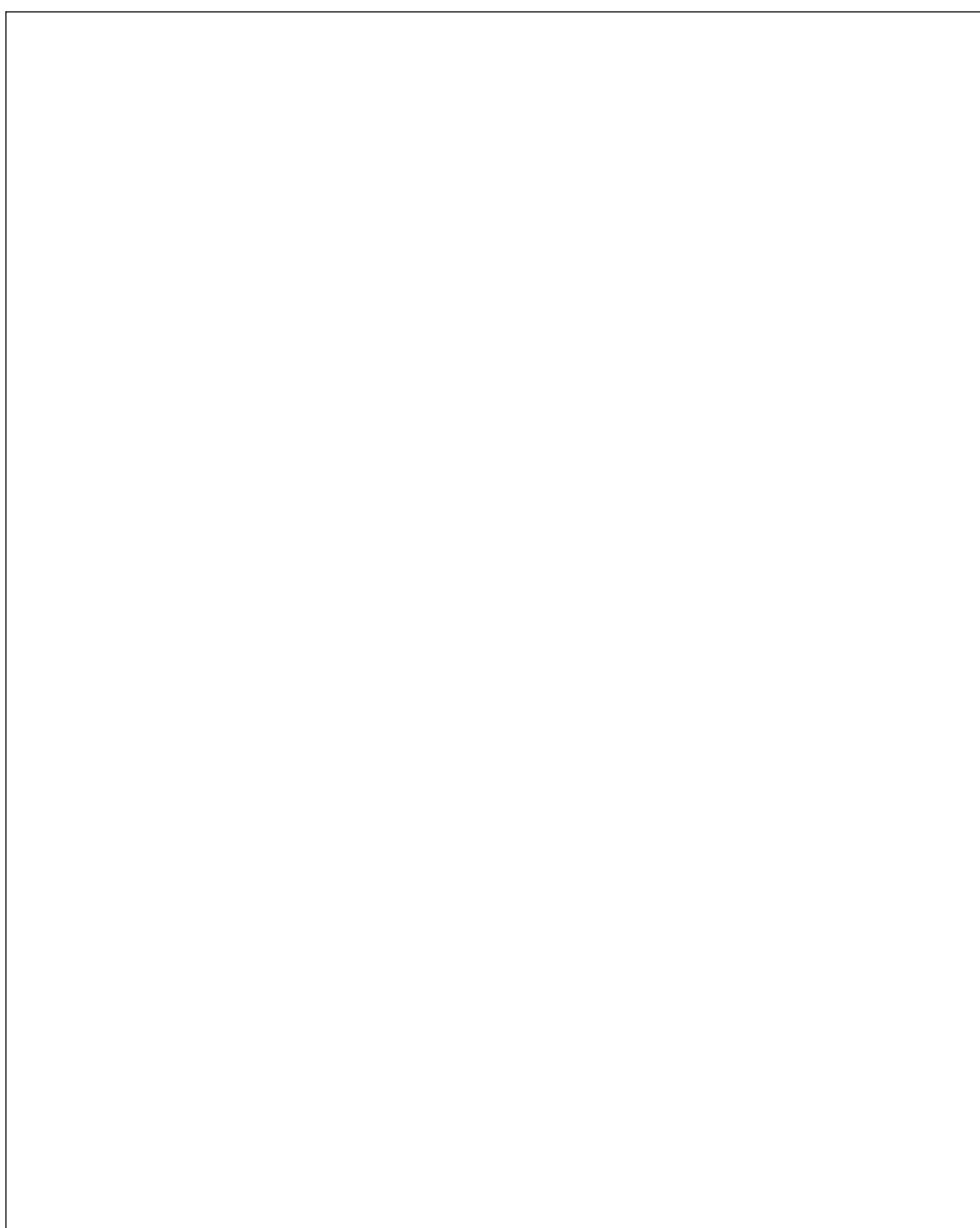
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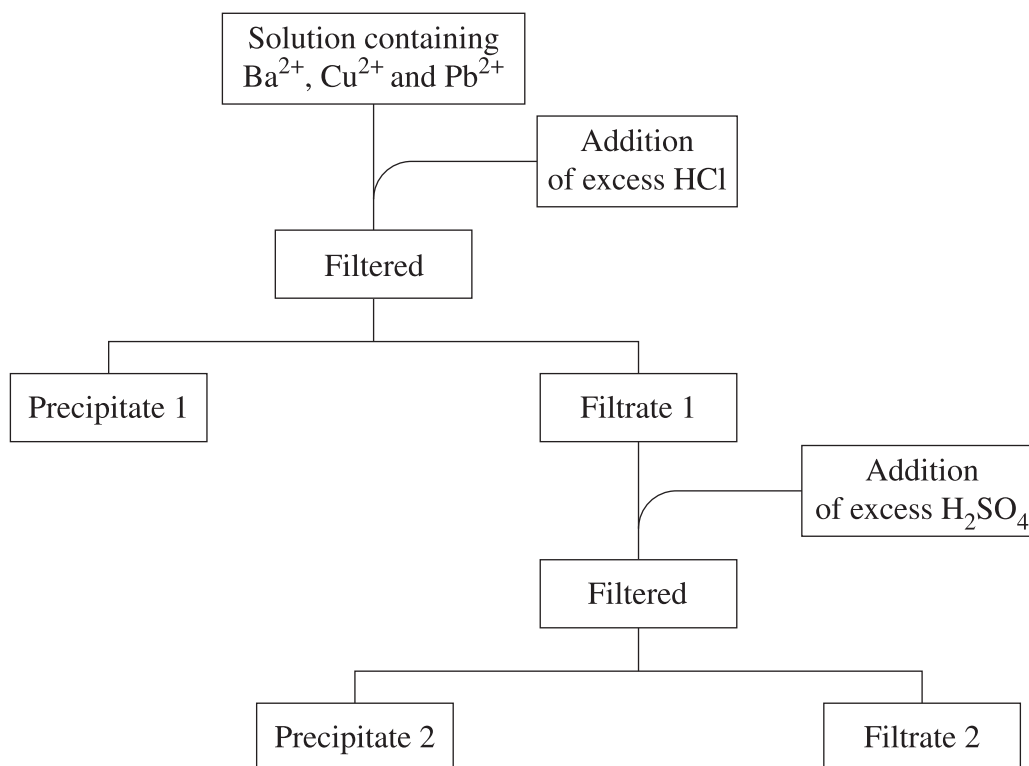
(b) Explain, on a labelled diagram, why reflux is used to produce an ester.

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Question 22 (5 marks)

A solution contains three cations, Ba^{2+} , Cu^{2+} and Pb^{2+} . The flow chart indicates the plan used to confirm the identity of these cations.



- (a) Name Precipitate 2. **1**

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- (b) Write a balanced net ionic equation for the formation of Precipitate 1. **2**

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- (c) Suggest a test and the expected result that would confirm the identity of the metal cation remaining in Filtrate 2. **2**

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Centre Number

Section I – Part B (continued)

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Student Number

Question 23 (7 marks)

A 20.72 g sample of solid lead was placed into 0.100 L of 1.00 mol L⁻¹ silver nitrate solution.

- (a) Complete the table. Show relevant calculations in the space below the table.

5

<i>Chemical species</i>	$\text{Pb}^{2+}(\text{aq})$	$\text{Pb}(\text{s})$	$\text{Ag}^{+}(\text{aq})$	$\text{Ag}(\text{s})$	$\text{NO}_3^{-}(\text{aq})$
<i>Moles in final mixture</i>					
<i>Balanced chemical equation</i>					

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- (b) With reference to only ONE species in the product mixture, explain why care must be taken in disposing of the final mixture.

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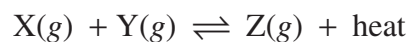
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Question 24 (4 marks)

Consider this chemical system which is at equilibrium.



- (a) Explain the effect of decreasing the volume of the reaction vessel. 2

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- (b) Explain the effect of adding a catalyst to this equilibrium mixture. 2

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Question 25 (4 marks)

An indicator is placed in water. The resulting solution contains the green ion, Ind^- , and the red molecule, HInd . 4

Explain why this solution can be used as an indicator. In your response, include a suitable chemical equation that uses Ind^- and HInd .

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Centre Number

Section I – Part B (continued)

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Student Number

Question 26 (4 marks)

Explain how microscopic membrane filters purify contaminated waters, in terms of their design and composition.

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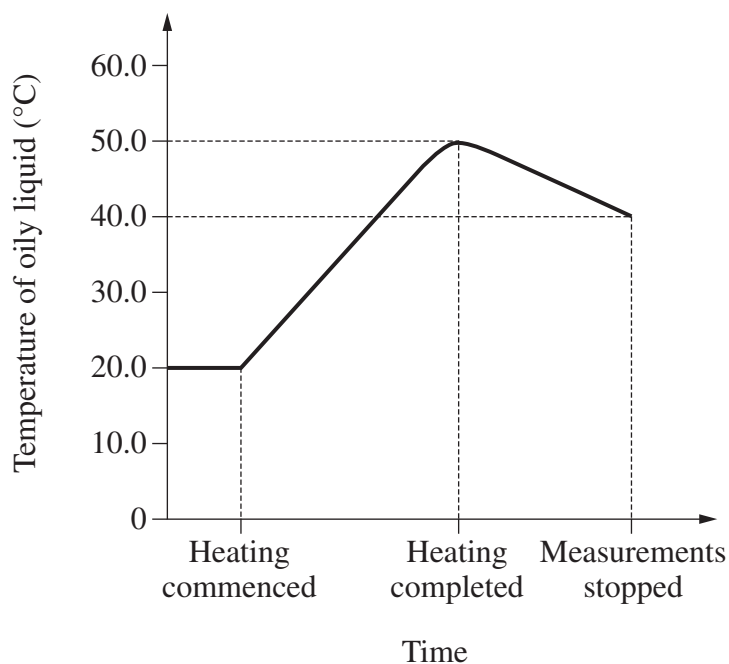
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Question 27 (4 marks)

A 0.259 g sample of ethanol is burnt to raise the temperature of 120 g of an oily liquid, as shown in the graph. There is no loss of heat to the surroundings.

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Using the information shown on the graph, calculate the specific heat capacity of the oily liquid. The heat of combustion of ethanol is 1367 kJ mol^{-1} .

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Centre Number

Section I – Part B (continued)

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Student Number

Question 28 (5 marks)

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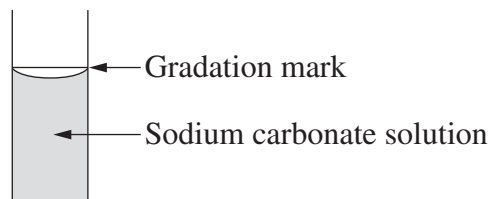
Question 28 (5 marks)

A student attempted to determine the concentration of a hydrochloric acid solution. The following steps were performed.

Step 1. A conical flask was rinsed with water.

Step 2. A 25.0 mL pipette was rinsed with water.

Step 3. The student filled the pipette with a standard sodium carbonate solution to the level shown in the diagram.



Step 4. The standard sodium carbonate solution in the pipette was transferred to the conical flask. The student ensured that all of the sodium carbonate solution was transferred to the conical flask by blowing through the pipette. Three drops of an appropriate indicator were added to the conical flask.

Step 5. A burette was rinsed with the hydrochloric acid solution and then filled with the acid. The student then carried out a titration to determine the concentration of the hydrochloric acid solution.

In steps 2, 3 and 4 above the student did not follow acceptable procedures.

- (a) Identify the mistake the student made in step 4 and propose a change that would improve the validity of the result.

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Question 28 continues on page 23

Question 28 (continued)

- (b) Explain the effect of the mistakes made in steps 2 and 3 on the calculation of the concentration of the hydrochloric acid solution.

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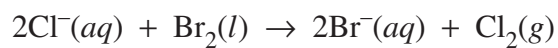
End of Question 28

Please turn over

Question 29 (2 marks)

Consider this chemical equation.

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Will the reaction occur spontaneously? Justify your response.

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Centre Number

Section I – Part B (continued)

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Student Number

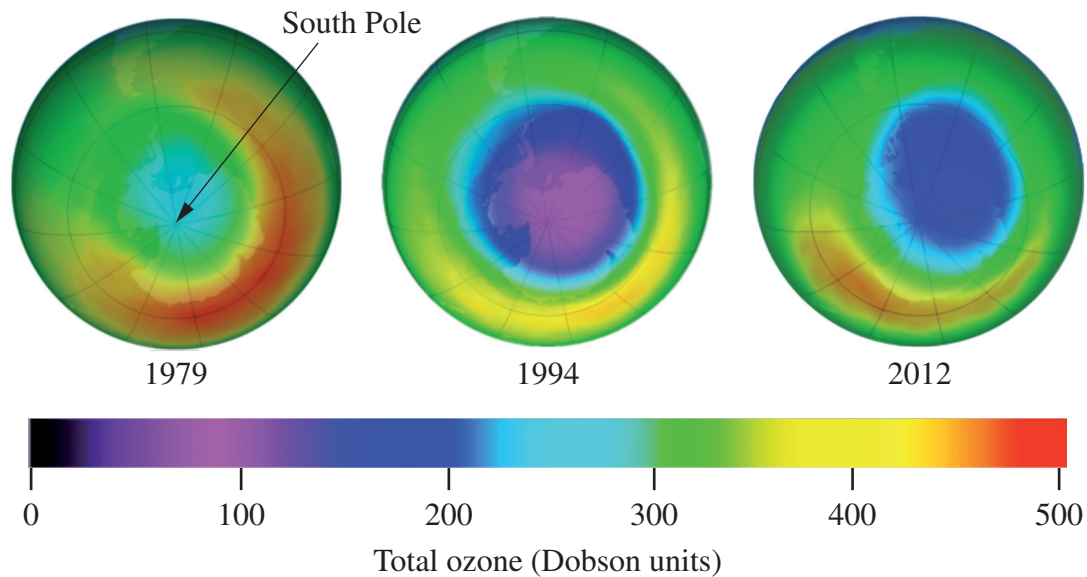
Question 30 (6 marks)

Please turn over

Question 30 (6 marks)

6

The images below represent the concentration of ozone in the atmosphere above Antarctica near the South Pole for the years 1979, 1994 and 2012, measured at the same time each year.



Acknowledgement: Courtesy of NASA

Account for the changes in ozone concentration above Antarctica between 1979 and 2012. Your response should include relevant equations.

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Question 30 continues on page 27

Question 30 (continued)

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End of Question 30

Please turn over

Question 31 (8 marks)

- (a) Construct separate flow diagrams to show the steps used in the production of polyethylene and those used in the production of a recently developed biopolymer.

5

Polyethylene	Name of biopolymer:
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- (b) Justify the use of a recently developed biopolymer over a polymer obtained from fossil fuel.

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Chemistry

Section II

25 marks**Attempt ONE question from Questions 32–36****Allow about 45 minutes for this section**

Answer parts (a)–(c) of the question in Section II Answer Booklet 1.

Answer parts (d)–(e) of the question in Section II Answer Booklet 2.

Extra writing booklets are available.

Show all relevant working in questions involving calculations.

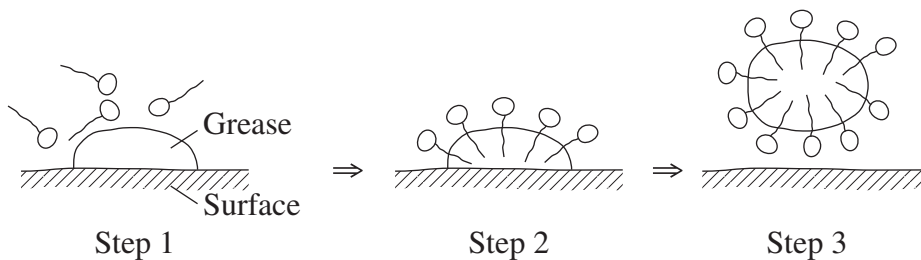
	Pages
Question 32 Industrial Chemistry	30–32
Question 33 Shipwrecks, Corrosion and Conservation	33–34
Question 34 The Biochemistry of Movement	35–38
Question 35 The Chemistry of Art	39–40
Question 36 Forensic Chemistry	41–44

Question 32 — Industrial Chemistry (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

- (a) The diagram shows a sequence of steps in the removal of grease from a surface. **3**

Explain the process shown in these steps.



- (b) Hydrogen iodide is a colourless gas that will decompose into colourless hydrogen gas and purple iodine gas according to the following endothermic reaction.



- (i) A 1.0 L glass container was filled with 0.60 moles of hydrogen iodide gas. When equilibrium was established, there were 0.25 moles of iodine gas present in the container. **3**

Calculate the equilibrium constant for this reaction.

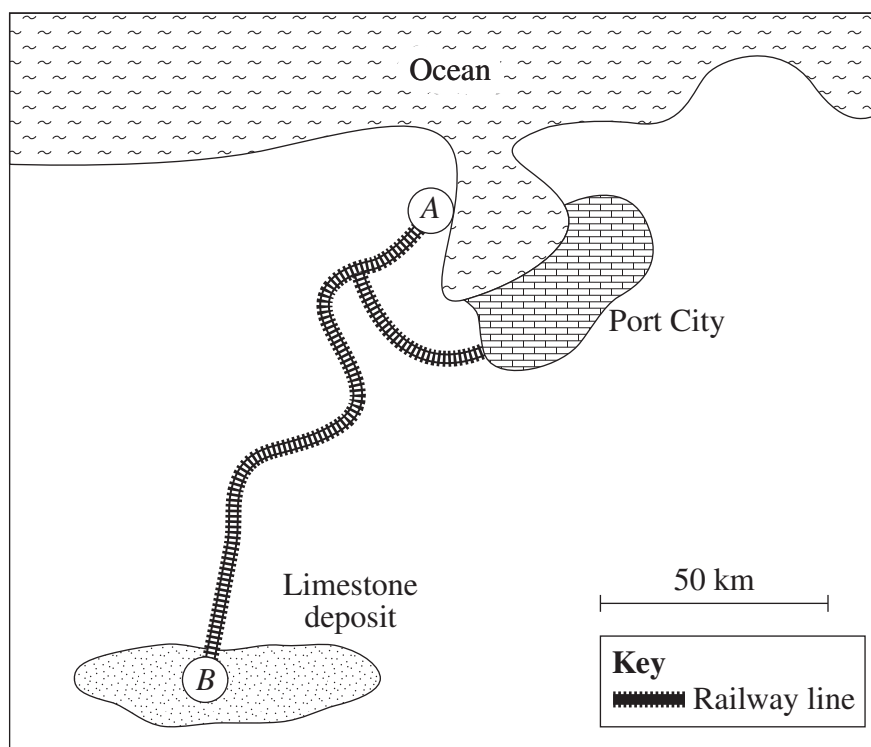
- (ii) The container was then cooled. **2**

Explain the change in the appearance of its contents.

Question 32 continues on page 31

Question 32 (continued)

- (c) The diagram shows a region where construction of a Solvay plant is being considered.



- (i) Why is limestone important in the Solvay process? Include a relevant chemical equation in your response. 2
- (ii) Justify which of sites A or B would be the preferred location for a Solvay plant. 3

Question 32 continues on page 32

Question 32 (continued)

Answer parts (d)–(e) in Section II Answer Booklet 2.

- (d) A first-hand investigation is conducted on the electrolysis of an aqueous solution of sodium chloride.
- (i) Justify the use of a safety precaution, other than wearing safety glasses, when carrying out this investigation. **2**
 - (ii) Describe how the THREE products of this electrolysis could be identified. **3**
- (e) There is often a compromise between maximising yield and minimising the environmental impact of industrial processes. **7**

Justify this statement with reference to the production of sulfuric acid.

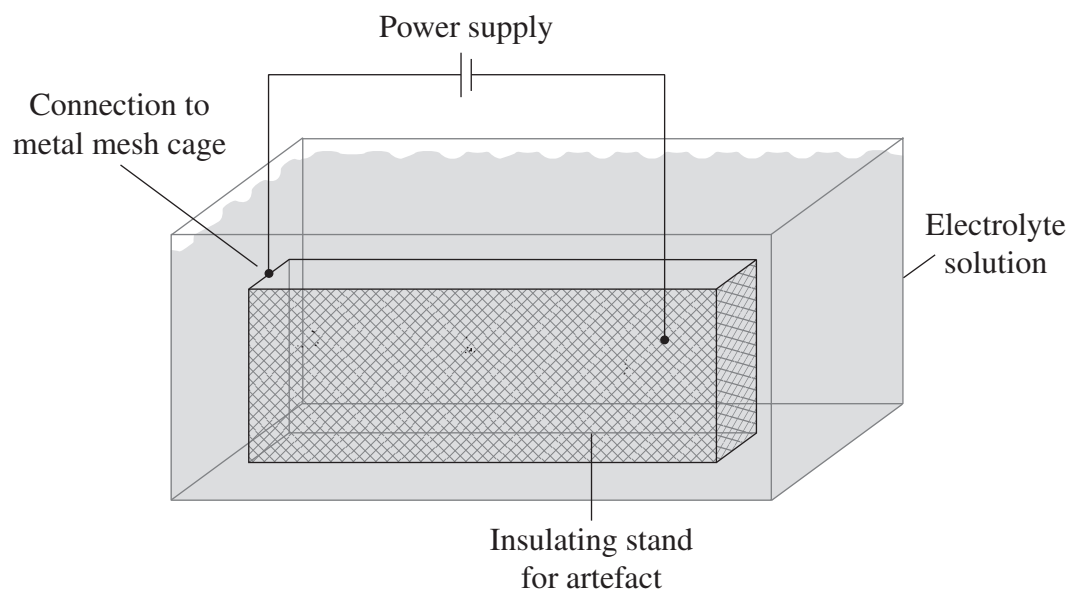
End of Question 32

Question 33 — Shipwrecks, Corrosion and Conservation (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

- (a) The treatment of a marine artefact is shown.

3



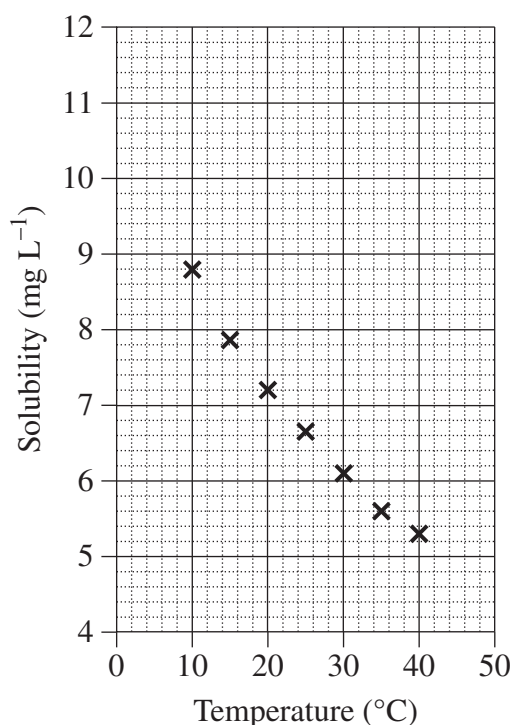
Explain why this process is used.

- (b) (i) Write a balanced chemical equation for the corrosion of tin in shallow water. **2**
- (ii) Summarise the role of electron transfer in corrosion reactions. Include relevant equations in your answer. **3**

Question 33 continues on page 34

Question 33 (continued)

- (c) The solubility of oxygen in sea water at various temperatures is shown in the graph.



- (i) Predict the solubility of oxygen at 4°C. 2
- (ii) Analyse the factors that affect the concentration of dissolved oxygen at increasing depths in the ocean. 3

Answer parts (d)–(e) in Section II Answer Booklet 2.

- (d) (i) Design a suitable first-hand investigation that could be carried out in a school laboratory to compare the effectiveness of different ways of preventing the corrosion of iron in a marine environment. Include relevant diagrams in your answer. 3
- (ii) Use the expected results of the investigation to justify the ongoing use of iron in the construction of ocean-going vessels. 2
- (e) From the scientists studied in this option, assess whose work has most changed the nature of scientific thinking about redox reactions. 7

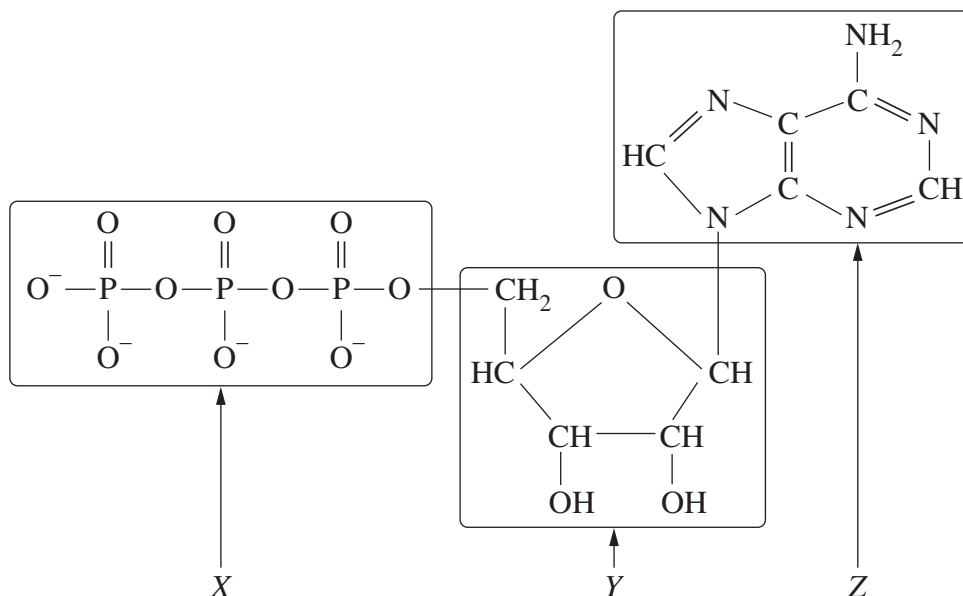
End of Question 33

Question 34 — The Biochemistry of Movement (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

(a) The molecule below is adenosine triphosphate (ATP).

3



Explain which part of the molecule, X, Y or Z, is the most important biologically. Include a relevant equation in your answer.

Question 34 continues on page 36

Question 34 (continued)

- (b) The table below gives the abbreviation and structural formulae for two amino acids.

Part of the glycine molecule has been replaced by X.

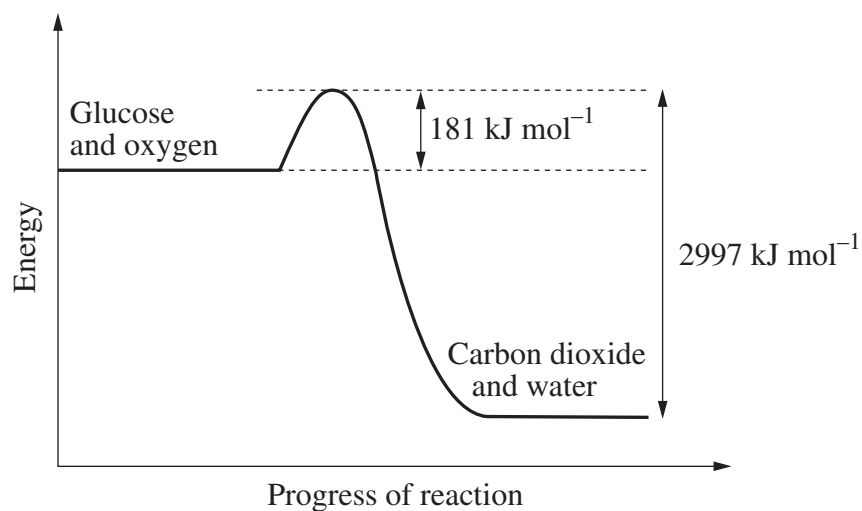
<i>Name</i>	<i>Abbreviation</i>	<i>Formula</i>
Glycine	Gly	$\begin{array}{c} \text{H} \\ \\ \text{X} - \text{C} - \text{COOH} \\ \\ \text{H} \end{array}$
Tyrosine	Tyr	$\text{HO} - \text{C}_6\text{H}_4 - \begin{array}{c} \text{H} \\ \\ \text{C} \\ \\ \text{H} \end{array} - \begin{array}{c} \text{H} \\ \\ \text{C} \\ \\ \text{NH}_2 \end{array} - \text{COOH}$

- (i) Write an equation using the correct structural formulae showing the formation of a dipeptide between glycine and tyrosine. 2
- (ii) Explain, in terms of chemical bonding, how the three-dimensional shape of a protein depends upon its primary structure. 3

Question 34 continues on page 37

Question 34 (continued)

- (c) The following energy profile is for the combustion of glucose during cellular respiration.



- (i) The hourly energy requirement of an astronaut is supplied by 36.0 g of glucose. 2

Using the energy profile above, determine the energy required by the astronaut in one hour.

- (ii) Calculate the volume of oxygen consumed by the astronaut in one hour. The conditions are 25°C and 100 kPa. Include a chemical equation in your answer. 3

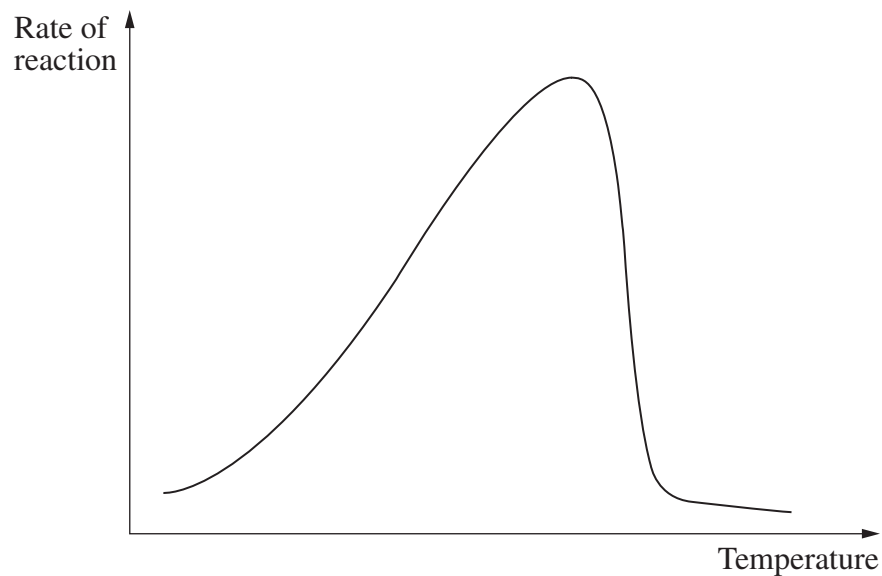
Question 34 continues on page 38

Question 34 (continued)

Answer parts (d)–(e) in Section II Answer Booklet 2.

- (d) A first-hand investigation was conducted to demonstrate the effect of temperature on the reaction of an enzyme.

- (i) Describe a procedure that could be used to demonstrate the effect of temperature on the reaction of an enzyme. **3**
- (ii) A sketch showing a student's results is provided below. **2**



Explain the shape of the curve.

- (e) 'Give me a muscle biopsy and I'll tell you whether you should be a sprinter or a marathon runner.' **7**

Evaluate the scientific accuracy of this statement in terms of muscle fibre structure and function.

End of Question 34

Question 35 — The Chemistry of Art (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

- (a) The emission spectrum of a metal salt is observed with a spectroscope.

3



Explain the processes by which emission lines arise. Include an energy level diagram for the lines marked **A** and **B** in your response.

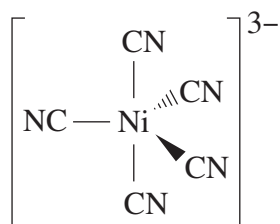
- (b) (i) Identify in which of the *s*, *p*, *d* or *f* blocks of the Periodic Table the element Ra is found. Justify your answer.

2

- (ii) An atom has FIVE valence electrons in its *d* orbital. Using an orbital diagram of the valence shell of the atom, explain how Hund's rule determines the electronic configuration.

3

- (c) The structure of a nickel cyanide complex is shown.



- (i) What is the oxidation state of nickel in this complex? Justify your answer.

2

- (ii) Explain the bonding of the complex with the aid of a Lewis structure for the cyanide ligand.

3

Question 35 continues on page 40

Question 35 (continued)

Answer parts (d)–(e) in Section II Answer Booklet 2.

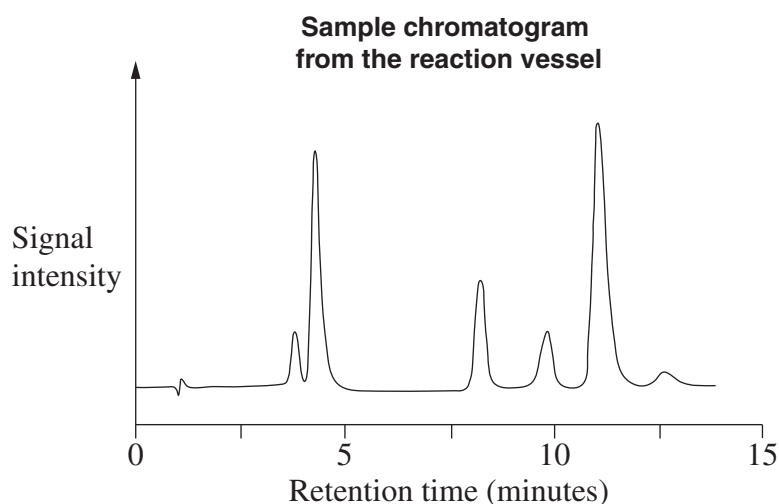
- (d) A first-hand investigation is conducted to demonstrate the oxidising strength of potassium permanganate.
- (i) Describe a procedure that could be used to demonstrate the oxidising strength of potassium permanganate. **2**
 - (ii) What does this investigation show about the oxidising strength of potassium permanganate? Use half-equations in your answer. **3**
- (e) Discuss the principles involved in using infra-red and ultraviolet spectroscopy for the analysis of pigments. **7**

End of Question 35

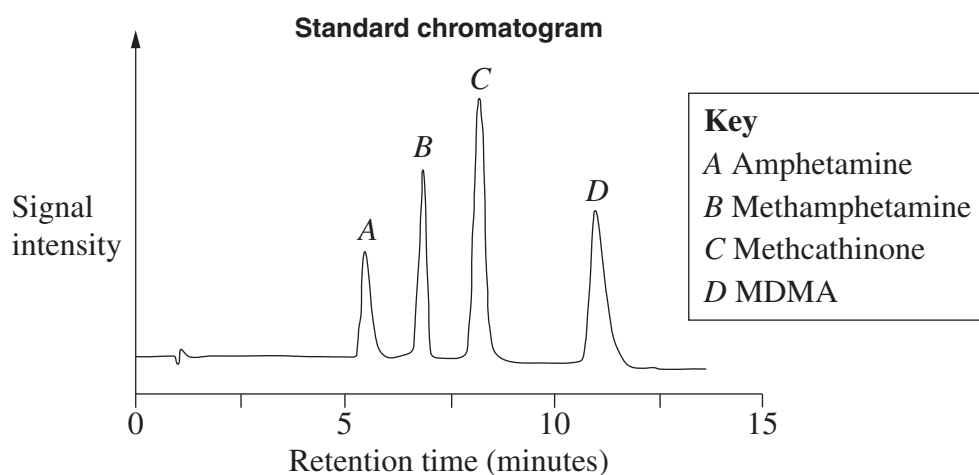
Question 36 — Forensic Chemistry (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

- (a) A sample was collected from inside a reaction vessel at a suspected illegal drug laboratory. The sample was analysed by high performance liquid chromatography (HPLC). The chromatogram obtained from the collected sample is provided below. **3**



The chromatogram of a standard mixture containing four compounds commonly produced in illegal drug laboratories is also provided.



By referring to the chromatograms, explain how this information could be used as evidence to support a guilty verdict in a court case.

Question 36 continues on page 42

Question 36 (continued)

- (b) (i) Information about three polysaccharides (cellulose, glycogen and starch) is provided in the table below. **2**

X	Y	Z
Plant polysaccharide	Animal polysaccharide	Plant polysaccharide
Contains branched and linear chains	Contains highly branched chains	Contains linear chains

Identify X, Y and Z.

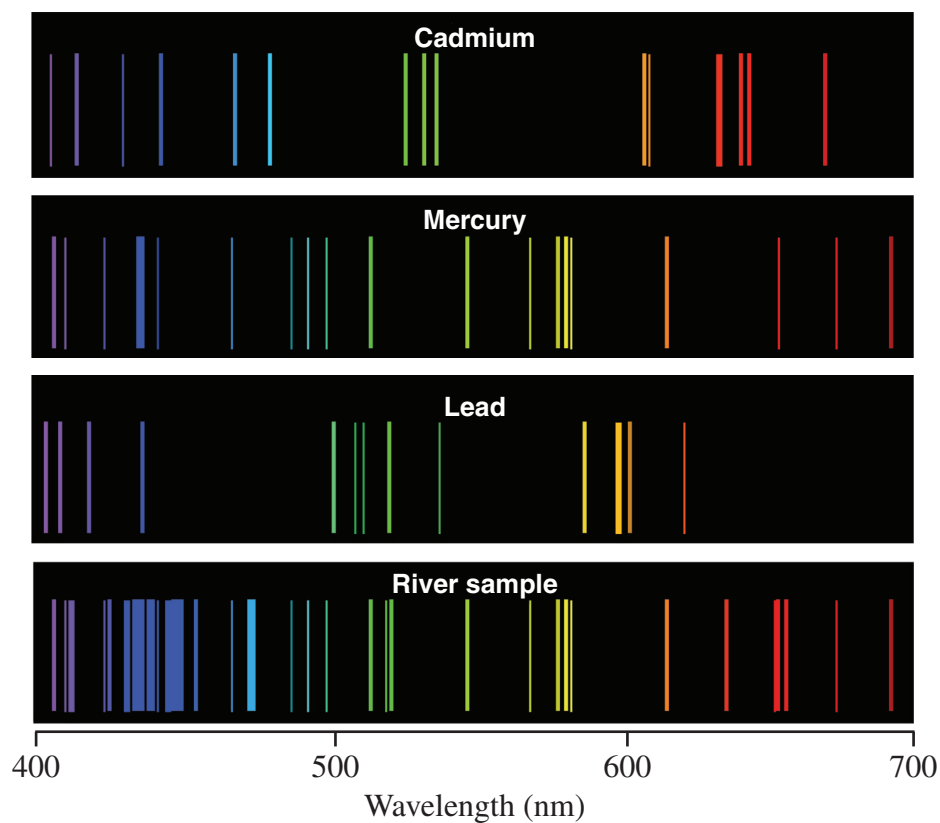
- (ii) Select ONE of cellulose, glycogen or starch and explain its primary function in terms of its composition and structure. **3**

Question 36 continues on page 43

Question 36 (continued)

- (c) A pesticide manufacturer is suspected of releasing waste water contaminated with heavy metal ions into a local river. Atomic emission spectroscopy is used to identify the possible pollutants.

- (i) Use the following emission spectra to identify a metal pollutant in the river. Justify your answer. 2



- (ii) Outline a procedure that could be used in a school laboratory to SAFELY produce and analyse the emission spectrum of an element. 3

Question 36 continues on page 44

Question 36 (continued)

Answer parts (d)–(e) in Section II Answer Booklet 2.

- (d) Alkanes, alkenes, alkanols and alkanoic acids are four different classes of organic compounds.
- (i) Describe a simple test that would confirm that a compound is organic. **2**
 - (ii) Describe a sequence of tests that could be used to distinguish between any **THREE** of the classes of organic compounds named above. **3**
- (e) A chemist working in a forensic laboratory is asked to analyse a residue of an explosive. They selected chromatography as the preferred analytical technique over electrophoresis. Justify this decision. **7**

End of paper

DATA SHEET

Avogadro constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 100 kPa and	
at 0°C (273.15 K)	22.71 L
at 25°C (298.15 K)	24.79 L
Ionisation constant for water at 25°C (298.15 K), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10}[\text{H}^+]$$

$$\Delta H = -m C \Delta T$$

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen		KEY																2 He 4.003 Helium																	
3 Li 6.941 Lithium		4 Be 9.012 Beryllium		Atomic Number Symbol Standard Atomic Weight Name																5 B 10.81 Boron		6 C 12.01 Carbon		7 N 14.01 Nitrogen		8 O 16.00 Oxygen		9 F 19.00 Fluorine		10 Ne 20.18 Neon					
11 Na 22.99 Sodium		12 Mg 24.31 Magnesium		79 Au 197.0 Gold																13 Al 26.98 Aluminium		14 Si 28.09 Silicon		15 P 30.97 Phosphorus		16 S 32.07 Sulfur		17 Cl 35.45 Chlorine		18 Ar 39.95 Argon					
19 K 39.10 Potassium		20 Ca 40.08 Calcium		21 Sc 44.96 Scandium		22 Ti 47.87 Titanium		23 V 50.94 Vanadium		24 Cr 52.00 Chromium		25 Mn 54.94 Manganese		26 Fe 55.85 Iron		27 Co 58.93 Cobalt		28 Ni 58.69 Nickel		29 Cu 63.55 Copper		30 Zn 65.38 Zinc		31 Ga 69.72 Gallium		32 Ge 72.64 Germanium		33 As 74.92 Arsenic		34 Se 78.96 Selenium		35 Br 79.90 Bromine		36 Kr 83.80 Krypton	
37 Rb 85.47 Rubidium		38 Sr 87.61 Strontium		39 Y 88.91 Yttrium		40 Zr 91.22 Zirconium		41 Nb 92.91 Niobium		42 Mo 95.96 Molybdenum		43 Tc Technetium		44 Ru 101.1 Ruthenium		45 Rh 102.9 Rhodium		46 Pd 106.4 Palladium		47 Ag 107.9 Silver		48 Cd 112.4 Cadmium		49 In 114.8 Indium		50 Sn 118.7 Tin		51 Sb 121.8 Antimony		52 Te 127.6 Tellurium		53 I 126.9 Iodine		54 Xe 131.3 Xenon	
55 Cs 132.9 Caesium		56 Ba 137.3 Barium		57–71 Lanthanoids		72 Hf 178.5 Hafnium		73 Ta 180.9 Tantalum		74 W 183.9 Tungsten		75 Re 186.2 Rhenium		76 Os 190.2 Osmium		77 Ir 192.2 Iridium		78 Pt 195.1 Platinum		79 Au 197.0 Gold		80 Hg 200.6 Mercury		81 Tl 204.4 Thallium		82 Pb 207.2 Lead		83 Bi 209.0 Bismuth		84 Po Polonium		85 At Astatine		86 Rn Radon	
87 Fr Francium		88 Ra Radium		89–103 Actinoids		104 Rf Rutherfordium		105 Db Dubnium		106 Sg Seaborgium		107 Bh Bohrium		108 Hs Hassium		109 Mt Meitnerium		110 Ds Darmstadtium		111 Rg Roentgenium		112 Cn Copernicium													

Lanthanoids

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.1 Ytterbium	71 Lu 175.0 Lutetium
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Actinoids

89 Ac Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium
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Elements with atomic numbers 113 and above have been reported but not fully authenticated.

Standard atomic weights are abridged to four significant figures.

Elements with no reported values in the table have no stable nuclides.

The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of data. Some data may have been modified.



2013 HSC Chemistry Marking Guidelines

Section I, Part A

Multiple-choice Answer Key

Question	Answer
1	B
2	C
3	D
4	C
5	A
6	A
7	C
8	A
9	A
10	C
11	B
12	B
13	D
14	B
15	C
16	B
17	B
18	D
19	A
20	D

Section I, Part B

Question 21 (a)

Criteria	Marks
Provides a reason for the use of acids as a food preservative related to properties of acids	2
Provides a reason for the use of acids	1

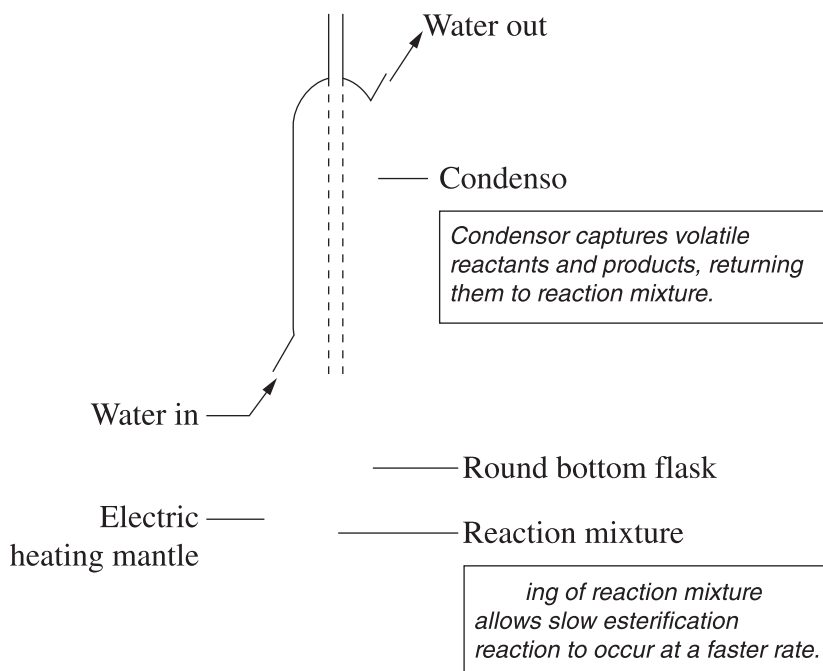
Sample answer:

Many microorganisms that cause spoilage of food cannot survive in low pH conditions. The acid added to a food prevents survival of these microorganisms by creating a low pH.

Question 21 (b)

Criteria	Marks
- Explains correctly the reasons for refluxing AND - Includes a well labelled correct diagram	4
- Describes a reason for refluxing AND - Includes a well labelled correct diagram	3
- Outlines a reason for refluxing AND - Includes a substantially correct diagram	2
- States a reason for refluxing OR - Includes a diagram recognisable as refluxing apparatus	1

Sample answer:



Question 22 (a)

Criteria	Marks
• Correctly identifies the name of Precipitate 2	1

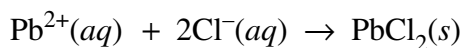
Sample answer:

Barium Sulfate.

Question 22 (b)

Criteria	Marks
• Provides a correct, balanced net ionic equation	2
• Provides a partially correct net ionic equation	1

Sample answer:


Question 22 (c)

Criteria	Marks
• Suggests a suitable test AND • Provides the observation expected	2
• Suggests a suitable test OR • Provides the observation expected	1

Sample answer:

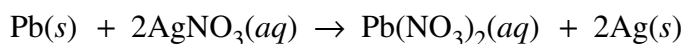
The remaining metal ion is Cu^{2+} which can be tested by a flame test. When present Cu^{2+} copper will turn the flame blue-green.

Answers could include:

- Tests for Pb^{2+} or Ba^{2+} if answers to (a) and (b) have been incorrect
- Other possible tests for Cu^{2+} , with their positive results

Question 23 (a)

Criteria	Marks
- Correct, balanced equation including solid states AND - All mole quantities correctly calculated AND - Calculation steps shown and correct significant figures used	5
- Substantially correct response with one minor error in: - Equation, or - Calculation, or - Significant figures	4
- Substantially correct equation - Attempts calculations with some working shown	2–3
Some relevant information provided	1

Sample answer:


$$n_{\text{Pb}} = \frac{20.72 \text{ g}}{207.2 \text{ g mol}^{-1}} = 0.1000 \text{ mol}$$

$$n_{\text{Ag}} = 0.100 \text{ L} \times 1.00 \text{ mol L}^{-1} = 0.100 \text{ mol}$$

Mole ratio Pb : AgNO₃ is 1 : 2

∴ 0.100 mol AgNO₃ and 0.0500 mol Pb used

Moles of remaining ions and solids

$$\text{Pb}^{2+}(aq) = 0.0500 \text{ mol}$$

$$\text{Ag}^{+}(aq) = 0 \text{ mol}$$

$$\text{NO}_3^{-}(aq) = 0.100 \text{ mol}$$

$$\text{Pb}(s) = 0.0500 \text{ mol}$$

$$\text{Ag}(s) = 0.100 \text{ mol}$$

Question 23 (b)

Criteria	Marks
- Identifies an appropriate species in the mixture AND - Relates the species to an appropriate potential health/environmental problem(s)	2
- Identifies an appropriate species in the mixture OR - Identifies potential health/environmental problem	1

Sample answer:

The reaction mixture contains lead, which is a heavy metal. Heavy metals are toxic to animals, including humans.

Answers could include:

Responses related to nitrates in mixture, with reference to environmental issues.

Question 24 (a)

Criteria	Marks
Correctly explains the effect of decreasing the volume of the reaction vessel	2
States an effect of decreasing the volume of the reaction vessel	1

Sample answer:

Decreasing the volume of the reaction vessel will increase the pressure. This will drive the reaction to the right hand side as there are fewer moles of gas on this side of the equation.

Question 24 (b)

Criteria	Marks
Correctly explains the effects of adding a catalyst to this reaction	2
- States ONE effect of adding a catalyst in general terms OR - States no overall change	1

Sample answer:

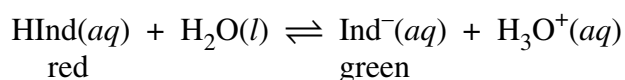
Adding a catalyst to the equilibrium mixture will have no visible effect on the reaction mixture. This is because a catalyst will speed up the rate of both the forward and the reverse reactions.

Question 25

Criteria	Marks
- Clearly articulates that indicators work through equilibrium processes, linking cause and effect through Le Chatelier's principle - Includes a chemical equation, correctly using the symbols given	4
- Some links made between the action of indicators and equilibrium reactions, with reference to the symbols given OR - Correct equations and use of symbols for an acid and base showing the action of the indicator without reference to equilibrium	3
Describes the action of indicators without the use of a chemical equation or reference to equilibrium	2
States one relevant concept	1

Sample answer:

An indicator needs to change colour in different pH conditions. The solution would have the equilibrium:



When a base is present, the concentration of H_3O^+ will be reduced. Le Chatelier's principle predicts the equilibrium will shift to the right, increasing the ionisation of the indicator. This shift causes the green colour to dominate.

Alternatively, when an acid is present the increased concentration of H_3O^+ will shift equilibrium left and the red colour will dominate.

Answers could include:

Separate equations to illustrate the response to an acid and a base.

Question 26

Criteria	Marks
Response includes a detailed description of the composition and design of microscopic membrane filters. Comprehensively links the purification of contaminated water to this composition and design	4
Response includes a description of the composition and design of microscopic membrane filters, and relates this to how these filters purify contaminated water	3
- Response includes a description of the composition and design of microscopic membrane filters OR - Includes a brief outline of the composition or design of microscopic membrane filters with reference to how they purify contaminated water	2
- Response gives a description of filtration OR - Response gives some features of the design or composition of membrane filters	1

Sample answer:

These filters consist of a thin film of polymer eg polypropylene, with tiny pores (typically about $0.2\ \mu\text{m}$ in size). The polymer is rolled into a thin, hollow tube and large numbers are bundled together to form a filtration unit. Dirty water flows along the outside of the tube, under pressure. Small particles, such as water molecules and ions, can pass through the pores while larger particles such as bacteria and protozoa are trapped by the membrane.

Answers could include:

Other feasible polymers. A labelled diagram.

Question 27

Criteria	Marks
- Calculates moles of ethanol AND - Calculates heat transferred AND - Correctly identifies temperature change from graph AND - Appropriately calculates specific heat capacity including units	4
Calculations substantially correct	3
At least ONE substantially correct calculation is given	2
Some relevant data is given	1

Sample answer:

$$n_{\text{C}_2\text{H}_5\text{OH}} = \frac{m}{M} = \frac{0.259 \text{ g}}{46.068 \text{ g mol}^{-1}} = 0.00562212 \text{ mol}$$

$$Q = \rho_{\text{H}_{\text{C}_2\text{H}_5\text{OH}}} \times n = 1367 \text{ kJ mol}^{-1} \times 0.00562212 \text{ mol}$$

$$= 7.68543 \text{ kJ} = 7685.43 \text{ J}$$

$$C = \frac{q}{m \times \Delta t}$$

$$= \frac{7685.43 \text{ J}}{0.120 \text{ kg} \times 30 \text{ K}} = \frac{7685.43 \text{ J}}{3.600 \text{ kg K}}$$

$$= 2134.84 \text{ J kg}^{-1} \text{ K}^{-1}$$

$$= 2.13 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$$

Question 28 (a)

Criteria	Marks
- Identifies the mistake made AND - Provides a proposed change that would improve the validity of the result	2
Identifies the mistake made	1

Sample answer:

Instead of blowing through the pipette, student should have touched the end of pipette to surface of flask to draw out the liquid.

Question 28 (b)

Criteria	Marks
Relates correctly the decreased number of moles caused by both steps to a decrease in volume of HCl to produce an overall increase in calculated concentration	3
- Identifies both mistakes and relates them to a reduction of moles OR - Relates reduced moles of one step to increased HCl concentration as calculated	2
- Identifies the reduced number of moles caused by either step OR - Identifies both mistakes	1

Sample answer:

In step 2, rinsing the pipette with water would decrease the number of moles of Na_2CO_3 it contains. This would be compounded by not filling to the mark made in step 3, resulting in the pipette having fewer moles of Na_2CO_3 .

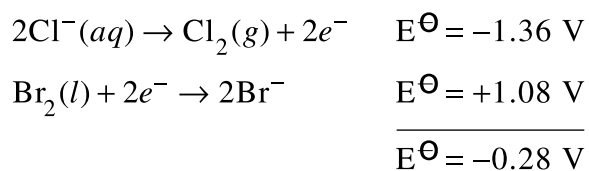
Hence, a lower volume of the HCl would be added from the burette, but the student would think that there were more moles of HCl in this volume. So, they will calculate the concentration of the acid solution to be higher than the actual value.

Question 29

Criteria	Marks
- Determines that the reaction will not be spontaneous AND (ONE of the following) - Refers to explicit $E^\ominus$ values from standard potential table OR - Justifies response using oxidation strength OR - Justifies response using position on the periodic table and electronegativities	2
- Determines that the reaction will not be spontaneous AND - Partially justifies response appropriately OR - Determines that the reaction will be spontaneous AND - Provides a logical justification	1

Sample answer:

No, the reaction will not occur spontaneously. With reference to the standard potential table:



The negative $E^\ominus$ suggests that the reaction will go from right to left spontaneously, not left to right.

Answers could include:

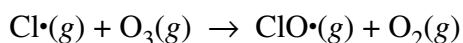
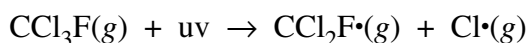
- Referral to chlorine being a stronger oxidant than bromine
- Referral to periodic table and chlorine being more electronegative than bromine

Question 30

Criteria	Marks
- Correctly identifies the decrease in ozone between 1979–1994 and accounts for the change - Includes relevant equations to show conversion of ozone to oxygen - Correctly identifies the increase in ozone between 1994–2012 and accounts for the change - Response should demonstrate coherency and present information in a logical way	6
- Correctly identifies the changes between 1979–1994 and 1994–2012 and accounts for the changes - Includes relevant equations for converting ozone to oxygen	4–5
Correctly identifies the changes between 1979–1994 and 1994–2012 with some attempt to account for the changes	2–3
Some relevant information provided	1

Sample answer:

The concentration of ozone above the South Pole decreased between 1979–1994. The decrease in ozone concentration was attributed to the prevalence of CFCs in the atmosphere. CFCs react (photo dissociate) to form chlorine radicals which are included in the conversion of ozone to molecular oxygen.



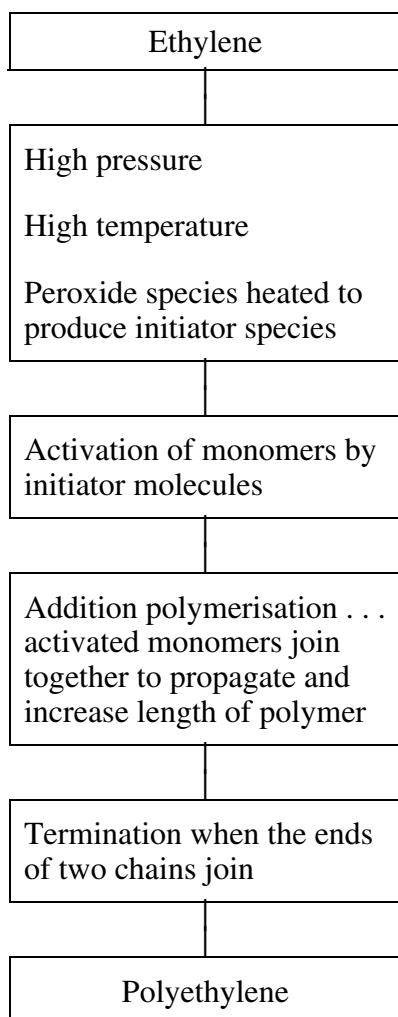
Since 1994 the concentration has begun to increase. A possible reason to explain this increase is the introduction of international agreements such as the Vienna Convention and the Montreal Protocol. These international agreements mandated a replacement of CFCs due to their ozone depletion potential (ODP). Alternatives to CFCs include HCFCs and HFCs which have a significantly lower ODP. The replacement of CFCs has resulted in a reduction in their atmospheric levels since the international agreements were enforced. This has seen the atmospheric concentration of ozone begin to increase, as shown by the 2012 ozone map.

Question 31 (a)

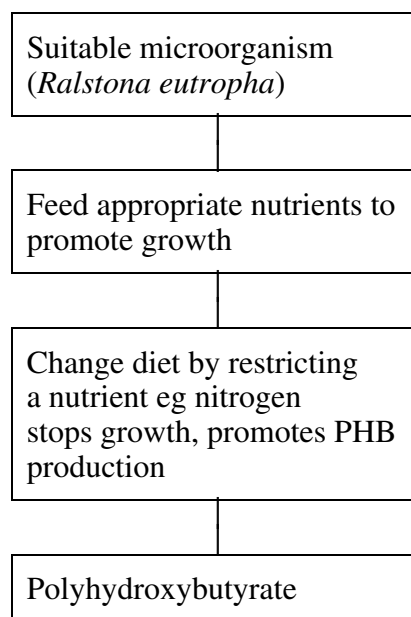
Criteria	Marks
• Provides correct processes for both materials	5
• Provides substantially correct series of steps for both materials	4
• Provides a correct series of steps for one material with all conditions OR • Provides a partially correct series of steps for both materials	3
• Provides a limited series of steps for one material OR • Provides relevant information for both materials	2
• Provides some relevant information	1

Sample answer:

Production of LDPE



Production of PHB



Answers could include:

- HDPE
- Other ways to produce biopolymer

Question 31 (b)

Criteria	Marks
- Links the uses of a recently developed biopolymer to its properties AND - Clearly uses this link to justify the use over fossil fuel polymers mentioning problems/disadvantages of the fossil fuel polymers	3
- Outlines properties of a recently developed biopolymer and gives one disadvantage or property of fossil fuel polymers OR - Links the use of recently developed biopolymer to some of its properties and attempts to justify the use of biopolymers over fossil fuel polymers	2
- Refers to a use or a property of a recently developed biopolymer OR - Refers to a fossil fuel polymer	1

Sample answer:

PHB (polyhydroxybutyrate) is a recently developed biopolymer used in the manufacture of medical supplies, nappies, plastic bottles and plastic packaging. The fact that it is biodegradable, biocompatible and a renewable resource produced from microorganisms distinguishes this biopolymer from the 'fossil fuel' polymers that have been traditionally used for these purposes to date. These fossil fuel polymers lack these properties and thus add significantly to problems such as non-biodegradable landfill and the accelerating use of non-renewable fossil fuel resources.

Answers could include:

There are other recently developed, named biopolymers.

Section II

Question 32 (a)

Criteria	Marks
Thorough explanation of the processes linking the structure of the surfactant to its ability to form an emulsion in water with grease	3
- Identifies the structure of the surfactant AND - Identifies the emulsifying property of the surfactant	2
- Identifies some features of the structure of the surfactant OR - Identifies that an emulsion is formed	1

Sample answer:

Soap has a hydrophilic, negatively charged polar head and a long, hydrophobic non-polar tail. When a mixture of soap and water is added the non-polar tails of the soap will dissolve in the grease. When enough soap has surrounded the grease, the particle is lifted off the surface. The negatively charged heads of the soap prevent individual grease droplets recombining, effectively forming an emulsion of grease in water, allowing the grease to be removed from the surface.

Answers could include:

Other surfactants instead of soap.

Question 32 (b) (i)

Criteria	Marks
Correct answer with relevant correct working shown	3
- Correct equilibrium expression with incorrect concentration values substituted in OR - Substantially correct equilibrium expression with correct concentration values substituted in	2
- At least one correct equilibrium concentration OR - Substantially correct equilibrium expression	1

Sample answer:

	HI	H ₂	I ₂
Initial	0.60	0	0
Used/made	0.50	0.25	0.25
Equilibrium	0.10	0.25	0.25

$$\begin{aligned}
 K &= \frac{[\text{H}_2][\text{I}_2]}{[\text{HI}]^2} \\
 &= \frac{0.25 \times 0.25}{(0.10)^2} \\
 &= 6.25
 \end{aligned}$$

Question 32 (b) (ii)

Criteria	Marks
Relates the cooling to equilibrium position to drive reaction to left and correctly indicates the purple colour due to I₂ will fade	2
- Statement that colour will fade OR - Statement that equilibrium will shift to left OR - Correct interpretation of incorrect statement	1

Sample answer:

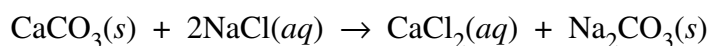
Since the reaction as written is endothermic, cooling the reaction vessel will drive the reaction to the left, reducing the concentration of purple I₂. Hence, contents of vessel will change to a lighter shade of purple.

Question 32 (c) (i)

Criteria	Marks
- Includes a correct reason for use of limestone as a source of CaCO_3 - Includes a correct equation	2
- States that limestone is a raw material OR - Substantially correct equation	1

Sample answer:

The major constituent of limestone is CaCO_3 which is a raw material for the Solvay process.


Answers could include:

Limestone heated to produce CO_2 that is used in the Solvay process.


Question 32 (c) (ii)

Criteria	Marks
Demonstrates relationships between several of the key requirements to appropriate locations and provides specific explicit reasons for a preferred location	3
Identifies some links between requirements and locations	2
States at least two requirements of the Solvay process	1

Sample answer:

Main requirements for the Solvay process include availability of raw materials (limestone and brine) and disposal of waste CaCl_2 , as well as some generalised requirements such as labour, access to markets.

While location *B* is close to the limestone deposit, it would be difficult to transport brine to *B*, and there is no easy way to dispose of waste. Location *A* would be preferred, as it is relatively easy to transport solid limestone to *A* by rail. Brine can easily be obtained from the ocean and waste CaCl_2 can be easily disposed of in the ocean. As well, *A* is close to the labour source of Port City, which would also have facilities to ship the product (Na_2CO_3) to markets.

Question 32 (d) (i)

Criteria	Marks
- Hazard identified related to products formed in experiment AND - A method of reducing the risk described	2
- Hazard identified OR - A method of reducing risk	1

Sample answer:

Chlorine gas is produced in this reaction which is toxic. The reaction must be carried out in the fume cupboard.

Answer could include:

Explosive mixture of H_2 and Cl_2

Electrical hazards

Question 32 (d) (ii)

Criteria	Marks
Includes details of all THREE products of electrolysis with descriptions of tests/observations to identify the products	3
Identification of TWO products of electrolysis with descriptions of tests/observations to identify the products	2
- Identification of at least TWO products of NaCl electrolysis OR - Description of a test to identify ONE product	1

Sample answer:

<i>Product of electrolysis of 4.0 mol L⁻¹ NaCl</i>	<i>How Identified</i>
Cl ₂ gas	- Bubbling at anode and - Universal indicator added was bleached colourless from green near anode by Cl₂ - Colourless I⁻ ions added were oxidised to brown I₂ molecules near anode by Cl₂(g). Presence of I₂ confirmed by adding starch (blue/black colour)
H ₂ gas	- Bubbling at cathode and - Gas collected pop test
OH ⁻ ions	Universal indicator changed from green to purple near cathode indicating OH⁻ ion present

Answer could include:

Description of identification of products for a dilute NaCl solution (H₂(g), OH⁻, O₂(g), H⁺).

Question 32 (e)

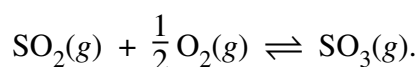
Criteria	Marks
- Coherent, concise response with no irrelevant extraneous information - Includes relevant equation - Thorough description of second step in process relating several factors to maximisation of yield - Clearly relates manipulation of reaction to reduction of environmental impact	7
- Includes relevant equation - Thorough description of second step in process relating several factors to maximisation of yield - Clearly relates manipulation of reaction to reduction of environmental impact	6
- Sound description of how several factors related to maximise yield are manipulated - Identifies that the process reduces environmental impact	4–5
- Process is described including some reaction conditions - An environmental issue identified	2–3
- One aspect of the production of sulfuric acid is identified OR - Environmental issue identified	1

Sample answer:

Maximising yield in the production of sulfuric acid occurs mainly in the second step of the contact process where sulfur dioxide is converted to sulfur trioxide.

The nature of this exothermic step requires a careful manipulation of the reaction conditions of concentration, pressure and temperature so that yield is maximised without comprising reaction rate. In addition, this manipulation also reduces the amount of SO₂ released into the atmosphere (a major environmental concern leading to formation of acid rain) and helps to dispose of waste heat produced in all steps in the process.

In the second, yield-determining step, the following reaction occurs:



To maximise yield, the initial air used contains about 30%–50% more oxygen than is required to fully convert sulfur to SO₃. This increased concentration of O₂(g) forces reaction to right, maximising yield. In addition, the reaction is carried out at about 1.5 atm. Since there is a 1.5 : 1 ratio of gas particles reactants : products the reaction is also forced to the right to reduce pressure, maximising yield.

The initial gas stream at the beginning of this step is at 1000°C after combustion of sulfur. This is too high to produce effective yield in the exothermic reaction above. The gas stream is cooled (heat exchanged, with heat used to liquefy sulfur in first step, which reduces heat released from process) to about 550°C.

At this lower temperature, the gas stream passes over a bed of V₂O₅ catalyst, which produces a relatively rapid 70% conversion of SO₂ to SO₃. The gas stream is then further cooled to

400°C and passed over a second catalytic bed, producing overall 97% conversion at the lower temperature. In the past, 3% SO₂ released was considered acceptable. However, now it is not permitted to release SO₂ at this concentration into the atmosphere.

So, after conversion of SO₃ to H₂SO₄ (via oleum), the remaining gas stream is passed over a final bed of V₂O₅ to produce a 99.7% conversion. 0.3% SO₂ released is considered acceptable.

Question 33 (a)

Criteria	Marks
Provides a clearly reasoned answer, identifying the treatment as electrolysis and the need to prevent further damage	3
- States ONE reason with supporting statement OR - States reasons for using the electrolysis	2
- States ONE reason for the electrolysis OR - Identifies the treatment as electrolysis	1

Sample answer:

This process would be used to remove salt from the artefact. This can be done because electrolysis can be used to remove chloride ions without damaging the artefact.

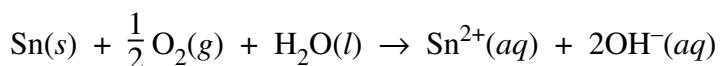
Answer could include:

- Reasons about cleaning and stabilising a metal artifact
- Correct half-equations/equations to support the answer

Question 33 (b) (i)

Criteria	Marks
Correctly balanced equation	2
Equation is substantially correct, with only minor errors	1

Sample answer:



Answer could include:

Other acceptable equations for the degradation of tin under those conditions.

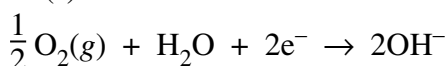
Question 33 (b) (ii)

Criteria	Marks
- Clearly identifies corrosion as a redox reaction - Provides half-equations to illustrate electron movement - Summarises the essential nature of the electron transfer	3
- Summarises the essential nature of the electron transfer without correct equations OR - Uses redox terms and equations, but fails to clearly articulate the need for electron transfer	2
- Identifies corrosion as a redox reaction OR - Provides a simple statement identifying the transfer of electrons in corrosion reaction	1

Sample answer:

Corrosion reactions are examples of redox reactions, where a metal will be oxidised and lose electrons. The transfer of electrons is essential to these reactions occurring as the oxidation and reduction processes need to be paired.

For example:



Without the movement of electrons corrosion would not occur.

Answer could include:

- Any suitable example to show the loss and gain of electrons
- Discussion of electrolyte present

Question 33 (c) (i)

Criteria	Marks
Answer between 10.0 and 11.8 mg L⁻¹ with units	2
- Answer between 9.2 and 12 mg L⁻¹ with units OR - Answer between 10.0 and 11.8 without units	1

Sample answer:

The solubility of oxygen at 4°C was found to be 11.0 mg L⁻¹.

Answer could include:

4°C → 11 mg L⁻¹.

Question 33 (c) (ii)

Criteria	Marks
- Outlines a range of factors which affect oxygen concentration in the ocean - Links the effects to different ocean depths	3
- Provides some information on a factor affecting oxygen concentration linked to ocean depth OR - Identifies a range of factors affecting oxygen concentration in oceans	2
Identifies a factor affecting oxygen concentration	1

Sample answer:

The factors affecting the concentration of dissolved oxygen at increasing ocean depths include:

- The temperature of the water which decreases at depth
- Currents of dense cold water with high oxygen concentrations at great depth
- Minimal mixing of surface water with water at intermediate depth
- Use of oxygen by organisms at different rates at different depths
- Little/no replacement of oxygen by photosynthesis due to lack of light at depth

Answer could include:

- References to anaerobic/aerobic bacteria
- More information on solubility of oxygen at the surface of the ocean

Question 33 (d) (i)

Criteria	Marks
- Experimental design includes a control and at least two methods of preventing corrosion - Treatment of variables clear, test is valid - Diagram included aids understanding	3
- Clear description of experimental design, with control, lacking a diagram OR - Includes diagram in description of experimental design, but lacking control or with an error in the treatment of variables	2
- Minimal description of experiment, without control or with errors in treatment of variables OR - Attempted diagram conveying some information about experimental procedure	1

Sample answer:

Experimental Set up

1. Control



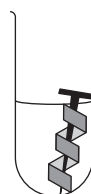
An untreated nail
in salt solution

2. Physical barrier



Nail painted,
allowed to dry
before placement
in salt solution

3. Chemical protection



Nail wrapped
with magnesium
ribbon

*NB: Equal volumes of same concentration solution in all test tubes.
All test tubes the same size.*

To test the effectiveness of corrosion prevention, a control is needed as shown above. The samples would need to be observed each day over a week-long period, identifying signs of rusting for each nail and drawing comparisons to that seen on the control.

Answers could include:

- Any specific examples of paints/resins/other coatings used to provide a physical barrier to protect the nail
- Any other metals added which would oxidize in place of the iron nail
- Applying a small current to the nail – provided the experimental set up appears feasible under school laboratory conditions
- More than three examples
- Any variations to equipment identified, provided a control is used and fair testing conditions are clear

Question 33 (d) (ii)

Criteria	Marks
• Provides reasons for using iron, making reference to preventing corrosion	2
• Provides some relevant information	1

Sample answer:

The expected results would show that iron can be protected by painting, galvanising or using a sacrificial anode. As these methods prevent corrosion it is reasonable to use iron in ocean-going vessels since it is also strong, relatively cost-effective and easily available.

Question 33 (e)

Criteria	Marks
- Coherent, concise response with no irrelevant, extraneous information - Proposes one of the four scientists studied - Justifies their choice with reference to the work of the other three OR the historical context of the research - Demonstrates an understanding of redox reactions - Discusses changes in scientific thinking	7
- Proposes one of the four scientists studied - Justifies their choice with reference to the work of the other three OR the historical context of the research - Demonstrates an understanding of redox reactions - Discusses changes in scientific thinking	6
- Proposes one of the four scientists studied - Justifies their choice with reference to EITHER the work of the other three OR the historical context of the research chosen - Identifies changes in scientific thinking - Demonstrates some understanding of redox reactions	4–5
- Proposes one of the four scientists studied AND - Makes some comparisons between the work of the scientists OR - Demonstrates an understanding of redox reactions through a description of the scientist's work	2–3
Suggests either Galvani, Volta, Davy or Faraday	1

Sample answer:

Our understanding of redox reactions has been the result of many scientists' work, including Galvani, Volta, Davy and Faraday. Arguably the work of Davy in identifying the connections between reactions and electricity most changed the direction or nature of scientific thinking in this area as his experiments were later used to develop the idea of electron transfer.

Prior to Davy, Galvani and Volta had worked on generating electric currents. While their work prompted further investigation, this continued along the direction of finding materials to generate electricity. Davy was able to recognise that the reactions were the source of the electric current and that the reactions involved decomposition. In this way he changed the direction of scientific thinking to electrolysis and was able to discover new elements (Mg, Ca, Ba).

Faraday was able to build on Davy's work and develop ways to quantify the relationship between the reactions and electricity in devising his laws of electrolysis. In this he continued and extended Davy's work, making very valuable contributions to our understanding, but not really changing the direction of scientific thinking.

While the term ‘redox reaction’ would not have been used at the time, the work of Davy has had the most significant impact on scientific thinking in this area, laying the foundations for qualitative and quantitative relationships between chemical reactions and electricity, later able to be identified as electron transfer reactions as more became known about structure and the nature of ionic substances.

Answers could include:

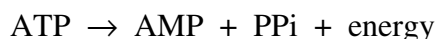
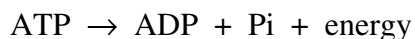
- Proposals for Galvani, Volta or Faraday which adequately argue for changes in scientific thinking resulting from their work

Question 34 (a)

Criteria	Marks
- Correctly identifies part X (phosphate groups) as the most biologically important AND - Relates hydrolysis of ATP (cleavage of phosphate groups) to the production of energy that is used by the organism AND - A relevant equation	3
- Correctly identifies part X AND - Relates hydrolysis of ATP to the production of energy OR - A relevant equation	2
- Correctly identifies part X OR - Relates hydrolysis of ATP to the production of energy OR - A relevant equation	1

Sample answer:

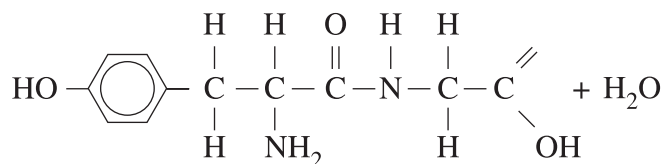
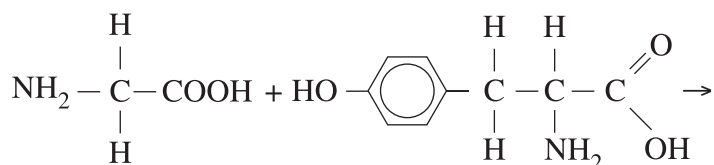
Part X is the most biologically important part of ATP (phosphate groups). ATP hydrolysis involves the breaking of the high energy bonds between phosphate groups which releases energy used primarily to ‘drive’ all other biochemical reactions. (It also produces metabolic heat that helps maintain body temperature.)



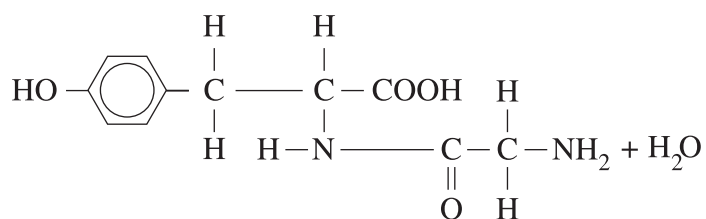
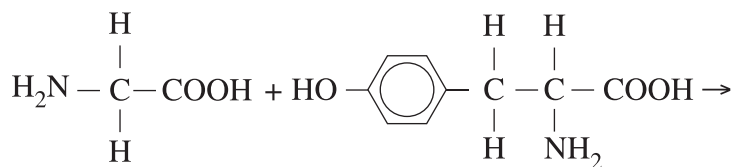
Question 34 (b) (i)

Criteria	Marks
Correctly draws the structural formulae including the missing amine functional group on glycine AND includes water as a product	2
- The structural formulae are correct except for the missing amine functional group OR - The amine group (on glycine) has been included	1

Sample answer:



Answers could include:



Question 34 (b) (ii)

Criteria	Marks
- Defines primary structure AND - Relates 3D shape to forces between side chains AND - States at least THREE of the forces between side chains	3
- Defines primary structure AND - States 1 or 2 of the forces between side chains OR - Relates shape to forces between side chains	2
- Defines primary structure OR - Mentions side chains OR - Names a force between side chains	1

Sample answer:

The primary structure of a protein is the sequence of amino acids that make up the protein. They are primarily linked by peptide bonds between carboxyl and amino functional groups.

However, the 3D shape of a protein is determined by side chains on the amino acid molecules. The sequence of amino acids will determine the side chains available to form subsidiary bonding between molecules.

Linkages include disulfide bridges, hydrogen bonding and electrostatic attractions which are all important in determining secondary shapes such as spirals, coils and sheets. The final 'tertiary' structure will be dependant upon these forces and any hydrophobic interactions caused by non-polar side chains.

Question 34 (c) (i)

Criteria	Marks
- Calculates moles of glucose correctly AND - Calculates energy correctly	2
- Calculates moles of glucose correctly but incorrectly calculates energy OR - Calculates moles incorrectly and there is no follow through error	1

Sample answer:

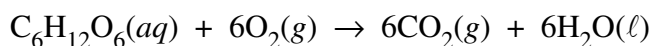
$$M_{\text{C}_6\text{H}_{12}\text{O}_6} = 180.156 \text{ g}$$

$$n_{\text{C}_6\text{H}_{12}\text{O}_6} \text{ required by the astronaut} = \frac{36.0 \text{ g}}{180.156 \text{ g mol}^{-1}} = 0.19982682 \text{ mol}$$

$$\begin{aligned} \text{Energy required (used) by astronaut} &= 0.19982682 \text{ mol} \times 2816 \text{ kJ mol}^{-1} \\ &= 562.712 \text{ kJ} \\ &= 563 \text{ kJ} \end{aligned}$$

Question 34 (c) (ii)

Criteria	Marks
- Equation for oxidation of glucose correct AND - Volume of oxygen correctly calculated	3
- One of the above OR - Equation AND moles oxygen incorrect but no follow through error	2
- Equation substantially correct OR - Moles of oxygen correct	1

Sample answer:


$$\begin{aligned} n_{\text{O}_2} \text{ consumed by astronaut} &= 6 \times n_{\text{C}_6\text{H}_{12}\text{O}_6} = 6 \times 0.1998 \text{ mol} \\ &= 1.1988 \text{ mol hr}^{-1} \end{aligned}$$

$$V_{\text{O}_2} = n_{\text{O}_2} \times 24.79 \text{ L mol}^{-1} = 1.19881 \text{ mol} \times 24.79 \text{ L mol}^{-1} = 29.7 \text{ L}$$

Question 34 (d) (i)

Criteria	Marks
Detail of controlled procedure (identifying or implying that temperature is the independent variable and identifying suitable dependent variable)	3
Some details included indicative of a controlled procedure	2
States some relevant information	1

Sample answer:

- Six separate water baths were prepared at different temperatures; 0°C, 20°C, 40°C, 60°C, 80°C and 100°C
- One test tube containing 5 mL of hydrogen peroxide (1%) was placed in each of the water baths, allowing the hydrogen peroxide substrate to reach the desired temperature
- Identical pieces of potato were used as a source of catalase. One piece of potato was put into each of the test tubes containing the hydrogen peroxide substrate at different temperatures
- The height of the column of oxygen gas (bubbles) consequent upon ‘mixing’ the potato and hydrogen peroxide substrate was measured after 5 minutes

Answers could include:

- Other catalase experiments using commercially obtainable catalase or other sources
- Rennin/milk investigation(s)

Question 34 (d) (ii)

Criteria	Marks
Relates the shape of the curve (peak, left/right of peak) to enzyme activity and denaturation	2
Relates part of the curve to appropriate enzyme activity	1

Sample answer:

The curve peaks at a particular temperature which corresponds to the optimal activity of the enzyme. Below this optimal temperature, molecule motion of both enzyme and substrate decreases resulting in fewer molecule collisions and less enzyme activity. This corresponds to the part of the curve to the left of the peak. To the right of the peak, the curve drops steeply, corresponding to enzyme denaturation above optimal temperature.

Question 34 (e)

Criteria	Marks
- Coherent, concise response with no irrelevant extraneous information - Clear judgement related to the scientific accuracy of the statement - Judgement based upon structure of fibre on biopsy - Relates this structure to slow or fast twitch metabolism - Relates fibre metabolism to suitability for sprinting or marathon running	7
- Clear judgement related to the scientific accuracy of the statement - Judgement based upon structure of fibre on biopsy - Relates this structure to slow or fast twitch metabolism - Relates fibre metabolism to suitability for sprinting or marathon running	6
- A judgement is given based upon structure of fibre and relates this structure to fast or slow twitch metabolism OR - Clearly relates fibre metabolism to suitability for sprinting or marathon running	4–5
- A judgement is given based on brief discussion of fibre types OR - Fibre metabolism is discussed	2–3
- A judgement is given OR - Different fibre types are mentioned	1

Sample answer:

The given statement is scientifically accurate. Skeletal muscle is comprised of different fibre types. It is the quantity and organisation of these in each of us that determines our genetic predisposition to a particular activity. A muscle biopsy would enable a prediction to be made with regard to that genetic predisposition.

Type I muscle fibres are red fibres with a good blood supply. They have a relatively good oxygen supply as well as a large supply of mitochondria and so they are well suited to aerobic respiration. This being the case, there is a plentiful, constant supply of ATP which is conducive to keeping these ‘slow twitch’ fibres contracting effectively over long periods of time. Biopsies of people with large proportions of these types of muscle fibres are consistent with people that are not particularly fast runners but can continue activities that require endurance, like marathon running. Enzyme profiles from these fibres are consistent with both carbohydrate and lipid metabolism over long time periods.

Type II muscle fibres are rapidly contracting ‘white fibres’ with a relatively poor blood supply. They have fewer mitochondria and that, in combination with less oxygen, leads to the fibres’ energy requirements being supplied largely through anaerobic respiration. Rapid glycolysis occurs with the formation of smaller numbers of ATP molecules, a metabolic pathway conducive to short, intensive bursts of activity such as sprinting.

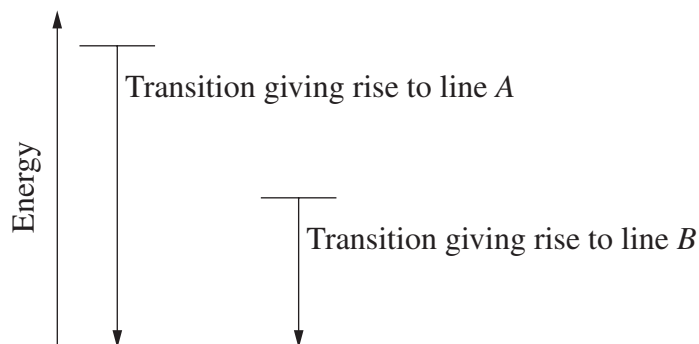
Muscle tissue biopsy can determine an individual’s overall percentage fibre makeup and could thus be used to determine that individual’s genetic suitability for a particular activity, sprinting or marathon running.

Question 35 (a)

Criteria	Marks
- Concepts of transition between discrete energy levels is articulated - Concept of emission lines being a transition from higher to lower energy level is articulated - An energy level diagram shows (approximately) correct relative size of transitions (Note: transitions do not need to end at the same lower energy state)	3
- A substantially correct energy level diagram - One other concept included	2
- One of the concepts is articulated OR - A substantially correct energy level diagram	1

Sample answer:

The metal ions in the solution gain energy from the components of the flame and are electronically excited, ie electrons are promoted to higher energy states. The electrons fall back to lower energy states, releasing energy as photons. The lines observed through the spectroscope are therefore emission lines. Because the electron energies are quantised, and transitions between states are restricted to well-defined energies, the energy released is well defined, thus the emission is seen as lines and not a continuous spectrum. Line *A* is the blue end of the spectrum, meaning the transition is of high energy (shorter wavelength) and line *B* is at the red end of the spectrum, meaning the transition is of low energy (longer wavelength).



Question 35 (b) (i)

Criteria	Marks
- Correctly identifies Ra as an s-block element - Appropriately justifies this by reference to valence shell electrons	2
Correctly identifies Ra as an s-block element	1

Sample answer:

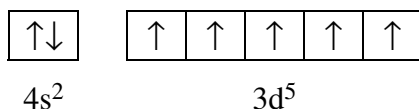
Ra (radium) is an s-block element. Its valence electrons are in the s orbital.

Answers could include:

Full electron configuration showing the valence shell as $7s^2$.

Question 35 (b) (ii)

Criteria	Marks
- Correct orbital diagram, showing correct occupancy of d sub-shell - Appropriate description of Hund's rule - Connection of orbital diagram to Hund's Rule	3
Two of above points	2
One of above points	1

Sample answer:


Hund's rule states that every orbital in a subshell is occupied with one electron before any orbital is doubly occupied. In this example Hund's rule is illustrated by the $3d$ sub-shell having 5 electrons in 5 different orbitals.

Answers could include:

Other orbital diagrams are possible, but must show a d sub-shell.

Question 35 (c) (i)

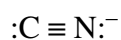
Criteria	Marks
- Identifies +2 as the oxidation state - Provides an appropriate justification	2
- Identifies +2 as the oxidation state OR - Provides an appropriate justification	1

Sample answer:

The oxidation state of nickel is +2. The overall charge of the complex is -3 , and each cyanide ion carries a -1 charge, giving a total of -5 from the cyanides. This means that the nickel must be +2 to give the overall charge of -3 .

Question 35 (c) (ii)

Criteria	Marks
- Includes a Lewis structure of cyanide that includes at least lone pair of electrons on carbon and negative charge - Appropriately includes description of bonding through carbon atom - Includes coordinate covalent OR Lewis acid/base bond term	3
Two of the above	2
One of the above	1

Sample answer:


The cyanide ion has lone pairs of electrons at both the carbon and nitrogen. The diagram indicates that the carbon of the cyanide ion forms a coordinate covalent bond through this lone pair to the nickel ion.

Answers could include:

Use of Lewis acid/base terminology rather than coordinate covalent bond.

Question 35 (d) (i)

Criteria	Marks
Provides a detailed procedure for a controlled investigation	2
- Outlines a procedure, lacking detail OR - Provides a procedure with variables confused	1

Sample answer:

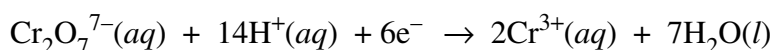
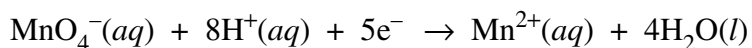
- Prepare identical concentrations of potassium permanganate, potassium chromate and iron(II) sulfate
- Add approximately 1 mL of iron(II) sulfate to a test tube
- Add up to 20 drops of potassium permanganate solution drop wise, tapping the side of the test tube to ensure mixing
- Record all observed colour changes
- Repeat for adding potassium permanganate to the potassium chromate

Question 35 (d) (ii)

Criteria	Marks
• Draws an appropriate conclusion • Has correct description of outcome of the investigation • Has substantially correct half-equations	3
• Draws an appropriate conclusion AND • Has correct description of the outcome of the investigation OR has substantially correct half-equations	2
• Draws an appropriate conclusion OR • Has correct description of the outcome of the investigation and conclusion OR • Has substantially correct half-equations	1

Sample answer:

Potassium permanganate solution was added drop wise to a solution of potassium chromate. The KMnO_4 solution decolourised as it was added, and the $\text{K}_2\text{Cr}_2\text{O}_7$ solution gradually changed from orange to green as the dichromate was reduced to Cr^{3+} . This shows that KMnO_4 is a stronger oxidiser than $\text{K}_2\text{Cr}_2\text{O}_7$.


Answers could include:

Any valid half-equation for other systems.

Question 35 (e)

Criteria	Marks
- Coherent, concise response with no irrelevant extraneous information - Includes relevant description of principles - Relates methods to analysis of pigments	7
- Includes relevant description of the principles - Relates methods to analysis of pigments	6
- Sound description of the principles - Identifies some factors related to analysis of pigments	4–5
- Basic description of the principles - Identifies some factors related to analysis of pigments	2–3
- Some principles given OR - Identifies some factors related to analysis of pigments	1

Sample answer:

Infrared and ultraviolet spectroscopies can be used to analyse pigments in paints. Both techniques rely on absorption of radiation to determine if a pigment is present, and where on the object the pigment is found. In both cases the absorption of the radiation results in the excitation of the pigment molecule to a higher energy state allowing for analysis.

Infrared and ultraviolet spectroscopies are commonly used in reflectance mode. The radiation penetrates the layers of paint. Infrared can penetrate the painting to the underlying base coat, while ultraviolet tends to reflect off the surface layer. Infrared is particularly sensitive to carbon-containing pigments such as charcoal and graphite, so it can be used to detect under-drawings or initial sketches.

The reflectance spectrum contains the radiation not absorbed by the pigments, and is therefore the complement of the absorption spectrum.

In general, infrared spectroscopy is a non-destructive method, especially when used in reflection mode, but some pigments may be heat sensitive and could change colour under intensive irradiation. Ultraviolet light may cause some degradation of pigments so should be used with caution.

Question 36 (a)

Criteria	Marks
- Identifies that illicit drugs (methcathinone – C and MDMA – D) are present in the sample mixture - Clearly explains how the information could be used	3
- Identifies that illicit drugs (methcathinone – C and MDMA – D) are present in the sample mixture - Outlines how the information could be used	2
Provides some relevant information	1

Sample answer:

Comparison between the standard and sample chromatograms indicates that the sample from the suspected drug laboratory contains Methcathinone and MDMA. This result confirms the drugs were being manufactured. Having established this result the forensic chemist may be requested to present this evidence in court. This will require the chemist to demonstrate the sample has not been contaminated and the chain of custody has been maintained at all times.

Answers could include:

Students may refer to the drugs as Drug C and Drug D.

Question 36 (b) (i)

Criteria	Marks
Correctly identifies all three polysaccharides	2
Correctly identifies one polysaccharide	1

Sample answer:

X – Starch

Y – Glycogen

Z – Cellulose

Question 36 (b) (ii)

Criteria	Marks
Identifies the main component of the selected polysaccharide and explains how its structure determines the main function	3
Describe how its structure (linear vs branched) determines the main function	2
Identifies the main function or structure	1

Sample answer:

Starch

Starch is a polysaccharide that contains a mixture of the branched amylopectin and unbranched amylose. The highly branched nature of amylopectin results in starch being readily broken down into glucose which represents an energy source. Due to the branched structure of amylopectin, starch is utilised as a storage polysaccharide by plants.

Answers could include:

Glycogen

Glycogen is a polysaccharide containing highly branched glucose units. The high levels of branching result in glycogen being readily broken down into glucose by various enzymes which is used as energy source. For this reason it is utilised as a storage polysaccharide in animals.

Cellulose

Cellulose is a polysaccharide formed by the linkage of the glucose units. The orientation of the units results in a linear chain, which allows other chains to closely pack with hydrogen bonds forming between linear chains. This increases the strength of cellulose and accounts for its low solubility and its function as a structural polysaccharide in plants.

Question 36 (c) (i)

Criteria	Marks
Correctly identifies mercury as the contaminant and appropriately justifies answer	2
- Correctly identifies mercury as the contaminant OR - Provides an appropriate justification	1

Sample answer:

It can be concluded that the river sample contains mercury because the spectral lines of the river sample match the spectral lines provided by mercury.

Question 36 (c) (ii)

Criteria	Marks
• Outlines an appropriate procedure AND • Identifies and addresses potential hazards	3
• Identifies an appropriate procedure and identifies a potential hazard OR • Correctly outlines an appropriate procedure without explicit identification of potential hazards	2
• Includes a brief, incomplete description of procedure OR • A suitable safety precaution	1

Sample answer:

A spectral tube containing the element was connected to a high voltage power supply. The power was switched on and the light produced was viewed through a spectroscope. The power supply must be switched off when connecting or disconnecting the spectral tube to avoid the potential for electrical shock.

Answers could include:

A flame test was carried out by heating a small sample of the metal salt in a blue Bunsen flame (Note: NOT mercury or lead if metal salt named). The coloured flame produced was viewed through a spectroscope. This experiment should be carried out in a fume cupboard.

(OR) Safety precautions could include taking care not to break the spectral tube if it contained mercury or lead.

Question 36 (d) (i)

Criteria	Marks
• Correctly describes a test for organic substances AND • States the positive result	2
• Identifies a test for organic substances	1

Sample answer:

If the substance burns in air to produce carbon, or carbon monoxide or carbon dioxide and water, it can be said to be organic.

Answers could include:

A simple test would be to combust the compound in plenty of oxygen. If it burns, producing carbon dioxide and water, it can be said to be organic.

Question 36 (d) (ii)

Criteria	Marks
Correctly produces a sequence of tests that could be used to distinguish between three of the named classes of compounds	3
Produces a sequence of tests that could distinguish between two of the named classes of compounds	2
States a test that could identify one of the classes of compounds	1

Sample answer:

Alkanoic acids, alkanes and alkenes are easily distinguished from each other. First, test the three samples with universal indicator. The alkanoic acid will turn the indicator red whereas the other two samples will be neutral and therefore green. To distinguish between an alkene and an alkane, add 10 drops of bromine water to a small sample of the compound. The alkene will rapidly decolourise the bromine water. The remaining sample will be the alkane.

Answers could include:

- Alkanoic acids can be distinguished from the other three classes by pH measurement
- Alkenes can be distinguished from the other three classes by the rapid decolourisation of bromine water
- Alkanols can be distinguished from alkanes by their reaction with a acidified potassium permanganate. The purple MnO_4^- is reduced to the colourless Mn^{2+} ion
- Alkanes can be identified by the fact that they will not react with any of the substances mentioned above.

Note: Answers may include a rice grain size piece of sodium metal to identify alkanols. Hydrogen gas is slowly produced. This would have been done as a demonstration.

Question 36 (e)

Criteria	Marks
- Coherent, concise response with no irrelevant or extraneous information - Demonstrates an extensive knowledge and understanding of both techniques - Presents a valid argument for why chromatography was selected - Includes advantages of chromatography with specific reference to its usefulness in forensic chemistry	7
- Demonstrates an extensive knowledge and understanding of both techniques - Presents a valid argument for why chromatography was selected - Includes advantages of chromatography with specific reference to its usefulness in forensic chemistry	6
- Demonstrates a sound knowledge and understanding of both separation techniques AND - Presents a valid argument for why chromatography was selected OR includes advantages of chromatography with some reference to its usefulness in forensic chemistry	4–5
Demonstrates a sound knowledge and understanding of separation techniques	2–3
Demonstrates a basic knowledge of a separation technique	1

Sample answer:

Chromatography is a separation technique based on the different solubilities of compounds in the stationary and the mobile phase. The stationary phase may be either a solid or liquid, while the mobile phase may be a liquid or a gas. The degree of solubility of a substance is determined by the polarity of the substance and the phases. Chromatographic separations are often improved by altering the mobile phase used which is typically a mixture of solvents. In contrast, electrophoretic separations are based on the charge and size of the components in the mixture. The pH of the electrophoresis solution (buffer) can be altered to change the charge on the substance to improve the separation. This differs from chromatography which relies on solubility differences in the various phases.

As chromatography has been selected as the preferred technique, it would indicate that the explosive residue components were not charged and therefore not suitable for analysis by electrophoresis.

Chromatography is an extremely useful technique in forensic chemistry as it can assist in identifying all individual components in a complex mixture found at a crime scene. This information can be used as collaborative evidence to link, or dismiss, a suspect to a crime scene. Chromatography is also a very sensitive technique which is useful for a forensic chemist as they often encounter very small sample sizes.

Chemistry

2013 HSC Examination Mapping Grid

Section I Part A

Question	Marks	Content	Syllabus outcomes
1	1	9.4.2.2.2	H8
2	1	9.4.3.2.2	H3, H6
3	1	9.4.4.2.1.0	H4
4	1	9.2.1.2.1	H10, H12.4b
5	1	9.2.5.3.1, 8.3.3.2.3	H6
6	1	9.3.2.2.1, 9.3.2.2.2	H6
7	1	9.4.1.2.3, 9.2.1.3.1	H9, H13
8	1	9.3.3.2.2	H9
9	1	9.2.1.2.5	H9
10	1	9.3.2.2.4	H8, H10
11	1	9.2.3.2.7	H7
12	1	9.2.4.3.1, 9.2.4.2.4	H7
13	1	9.2.2.2.4	H4, H9
14	1	9.3.3.2.5	H10
15	1	9.3.2.2.9	H10, H12.4b
16	1	9.4.4.2.4	H6
17	1	9.3.3.2.2, 9.3.3.2.6	H10
18	1	9.3.5.2.2, 9.3.5.2.4, 9.3.5.2.5, 9.2.1.2.2, 9.2.1.2.3, 9.2.3.2.2	H9
19	1	9.3.1.3.1, 9.3.1.2.2, 9.3.4.2.8	H10, H11
20	1	9.4.4.2.6	H6

Section I Part B

Question	Marks	Content	Syllabus outcomes
21 (a)	2	9.3.3.3.5	H4
21 (b)	4	9.3.5.2.6, 9.3.5.3.1	H8, H11.3a, H13
22 (a)	1	9.4.3.2.1, 9.4.3.2.1	H6, H14
22 (b)	2	9.4.3.2.1, 9.2.1.3.1	H10, H14
22 (c)	2	9.4.3.2.1, 9.4.3.3.1	H6, H14
23 (a)	5	9.2.4.2.2, 9.4.5.2.1, 9.4.5.3.1, 8.3.4.3.3, 8.4.4.3.4	H10
23 (b)	2	9.4.3.3.2, 9.4.5.3.2	H4, H11.3b, H12.1c,d
24 (a)	2	9.3.2.2.4, 9.4.2.2.9	H8
24 (b)	2	9.3.2.2.4	H8
25	4	9.3.1.3.1, 9.3.4.2.3, 9.3.4.2.5, 9.3.2.2.4	H8, H13.1d, H14.1g

Question	Marks	Content	Syllabus outcomes
26	4	9.4.5.2.4	H3, H13
27	4	9.2.3.2.7	H7, H10, H13
28 (a)	2	9.3.4.2.8	H11.2c, H11.2d, H12.1a
28 (b)	3	9.3.4.2.8, 9.3.4.3.3	H2, H12.1a, H12.4a
29	2	9.2.4.3.4	H7, H12.3c, H14.1d
30	6	9.4.4.3.1, 9.4.4.3.3, 9.4.4.2.10-11	H4, H8, H13.1e
31 (a)	5	9.2.1.2.6, 9.2.2.3.1	H3, H9, H13
31 (b)	3	9.2.2.3.1	H4, H9

Section II

Question	Marks	Content	Syllabus outcomes
Question 32		Industrial Chemistry	
(a)	3	9.5.5.2.3, 9.5.5.5.4	H8, H9
(b) (i)	3	9.5.2.2.2, 9.5.2.3.3	H10, H13.1c, H13.1d
(b) (ii)	2	9.5.2.2.1, 9.5.2.2.3	H7, H8
(c) (i)	2	9.5.6.2.1, 9.5.6.2.3	H8, H10
(c) (ii)	3	9.5.6.3.3, 9.5.6.2.4	H14.1b, H14.1c, H14.1h
(d) (i)	2	9.5.4.3.1	H11.2e, H11.3b, H12.1d
(d) (ii)	3	9.5.4.3.1, 9.5.4.3.2	H7
(e)	7	9.5.3	H2, H3, H4, H8, H10
Question 33		Shipwrecks, Corrosion and Conservation	
(a)	3	9.6.7.2.3, 9.6.7.2.4	H7, H8
(b) (i)	2	9.6.1, 9.6.3	H10, H13.1d
(b) (ii)	3	9.6.1.2.2	H8
(c) (i)	2	9.6.5.2.1	H8, H14.1
(c) (ii)	3	9.6.5.2.3	H8, H14.1e
(d) (i)	3	9.6.4.3.3	H8, H11.1e, H11.2b, H11.3a, H13.1e
(d) (ii)	2	9.6.4.3.1, 9.6.2.2.2	H3, H8, H14.1c
(e)	7	9.6.1.3.1, 9.6.1.2.4	H1, H2, H7, H8, H13.1a, H14.1h, H14.3b

Question 34		The Biochemistry of movement	
(a)	3	9.7.1.2.2, 9.7.1.3.1	H7, H9
(b) (i)	2	9.7.4.2.2, 9.7.4.2.3	H10, H13.1d, H13.1e
(b) (ii)	3	9.7.4.2.5	H9
(c) (i)	2	9.7.7.8.1, 9.7.8.3.2	H9, H10, H12.3c, H12.4b, H13.1d
(c) (ii)	3	9.7.8.2.5	H12.3c, H12.4b, H13.1d
(d) (i)	3	9.7.4.3.2	H9, H11.1b, H11.2a-b-c
(d) (ii)	2	9.7.4.3.2, 9.7.4.2.6	H8, H9
(e)	7	9.7.5.3.1, 9.7.8, 9.7.10	H2, H7, H8
Question 35		The Chemistry of Art	
(a)	3	9.8.2.2.3, 9.8.2.2.7	H6, H7, H13.1e
(b) (i)	2	9.8.3.2.6	H6
(b) (ii)	3	9.8.3.3.2, 9.8.4.3.1	H6, H13.1e
(c) (i)	2	9.8.5.2.3, 9.8.4.3.3	H6
(c) (ii)	3	9.8.5.2.4, 9.8.5.3.1	H6, H8, H13.1d, H13.1e
(d) (i)	2	9.8.4.3.4, 9.8.4.2.4, 9.8.4.3.2	H8, H11.1b, H11.2c
(d) (ii)	3	9.8.4.3.4, 9.8.4.2.4, 9.8.4.3.3	H8, H14.1a-b-d-e
(e)	7	9.8.2.2.4, 9.8.2.3.3, 9.8.2.2.9	H3, H6, H11.1a, H11.3a
Question 36		Forensic Chemistry	
(a)	3	9.9.5.2.2, 9.9.5.3.1	H9
(b) (i)	2	9.9.2.2.2, 9.9.2.2.4	H9, H14.1a
(b) (ii)	3	9.9.2.2.4, 9.9.2.3.2	H9, H14.1h
(c) (i)	2	9.9.6.3.2, 9.9.6.2.5	H6, H14.1b
(c) (ii)	3	9.9.6.3.1, 9.9.6.2.5	H6, H11.3a-b, H11.2c-e
(d) (i)	2	9.9.1.2.2, 9.9.1.3.3	H8, H9
(d) (ii)	3	9.9.1.2.3, 9.9.1.3.3	H8, H9, H11.2d
(e)	7	9.9.3.2.5, 9.9.3.2.6, 9.9.5.2.2, 9.9.5.3.1	H3, H4, H9, H11.3a, H11.3c H14.2a-b, H14.3b